Long-Term Risk Differentiation

Rodrigo Peña*

CEMFI

November 10, 2026

Abstract

Temporary fluctuations in borrowers' financial conditions can change their credit ratings even when there are no changes in persistent creditworthiness. Under the Internal Ratings-Based (IRB) framework, calibrating probabilities of default to long-run experience does not correct rating assignments driven by these fluctuations. This paper quantifies differences between ratings based on current and persistent creditworthiness and examines whether incorporating the latter improves long-term default-risk differentiation. Using Bank of Spain credit scores for Spanish firms over 2000–2023, I estimate a state-space model separating persistent firm-specific creditworthiness from aggregate movements and transitory firm-specific variation. Smoothed estimates provide a benchmark, while filtered estimates yield ratings based on current and past scores. Point-in-time (PIT) ratings differ from the benchmark in 51.2% of firm-year observations, compared with 27.9% for filtered ratings. Filtered ratings increasingly outperform PIT ratings in differentiating future defaults as the horizon lengthens for medium, small, and micro firms, but not for large firms. Neither mechanical smoothing nor removing aggregate movements reproduces these gains, highlighting the importance of firm-specific transitory variation.

Keywords: Credit ratings; Internal rating systems; Long-term creditworthiness; Regulatory capital; Risk differentiation.

*I am deeply grateful to Enrique Sentana for his guidance and advice. I also thank Dante Amengual, Manuel Arellano, Rubén García Céspedes, Javier Mencía, Melissa Prado, Rafael Repullo, Víctor Sancibrián, Anatoli Segura, and Rafael Zambrana for their valuable comments and suggestions. I am grateful to the participants in the PhD Mentoring Day at the 33rd Finance Forum. I also thank Sergio Gavilá, Lola Morales, and Rafael Vivo for their support and helpful feedback during my internship at Banco de España and throughout the subsequent development of this paper. Finally, I gratefully acknowledge financial support from the Santander Research Chair at CEMFI and Fundación Ramón Areces.

1 Introduction

Credit ratings play a central role in credit markets, supporting decisions ranging from lending and pricing to regulatory capital requirements. These different applications, however, do not require the same notion of creditworthiness. Short term lending and pricing decisions benefit from ratings that respond quickly to new information, whereas applications concerned with long-term solvency, like bank capital requirements determination, require ratings that abstract from temporary fluctuations. This distinction is commonly reflected in the use of Point-in-Time (PIT) and Through-the-Cycle (TTC) ratings. PIT ratings respond to recent firm-specific and macroeconomic developments, whereas TTC ratings seek to isolate borrowers' long-term creditworthiness from transitory and cyclical movements.

Under the Internal Ratings-Based (IRB) approach, banks estimate probabilities of default (PDs) that reflect long-run average one-year default rates while maintaining internally consistent rating systems for both risk management and regulatory capital purposes.[1] For analytical purposes, the design of a rating system can be separated into two closely related stages:[2]

- *Risk differentiation*: Borrowers are ranked according to their relative default risk, and the resulting assessments inform their assignment to rating grades.
- *Risk quantification*: PDs attached to those grades are calibrated to reflect long-run default experience.

Importantly, calibrating PDs to long-run average default rates does not imply that the underlying rating assignments reflect persistent creditworthiness. Although the regulatory framework requires rating assignments to consider risk beyond a one-year horizon, it does not prescribe a unique PIT or TTC rating philosophy. In particular, rating methods that are sensitive to current conditions can coexist with PD calibration to long-run average default rates.[3] However, temporary firm-specific shocks and heterogeneous exposure to aggregate conditions may affect borrowers' observed scores even when there are no changes in their persistent

[1] Basel Committee on Banking Supervision (2006), paras. 444 and 447; Regulation (EU) No 575/2013, Arts. 144(1)(b) and 180(1)(a).

[2] See EBA/GL/2017/16, *Background and rationale*, particularly the discussion of model development and calibration and Figure 3; see also EBA (2023), *Supervisory Handbook on the Validation of IRB Rating Systems*, para. 29.

[3] See Regulation (EU) No 575/2013, Art. 171(3), as amended by Regulation (EU) 2024/1623, and Basel Framework, CRE36.29–36.30, on the rating-assignment horizon. EBA/GL/2017/16, *Background and rationale*, pp. 23–25 and Table 1, explicitly discusses PIT-sensitive ranking combined with TTC calibration. For evidence that this combination occurred in surveyed models, see EBA (2017), *Report on IRB Modelling Practices*, Section 4.4, para. 80.

creditworthiness. Such fluctuations can alter grade assignments and may even reverse the relative ordering of borrowers according to their persistent credit quality. Recalibrating the PDs attached to existing grades without changing the underlying assignments cannot correct these differences.

This paper makes two contributions. First, it quantifies the differences in borrower classification that arise when risk differentiation reflects current rather than persistent creditworthiness. Second, it proposes and evaluates a method for incorporating persistent creditworthiness into rating assignments using current and past credit scores. Both contributions use a state-space framework that separates credit-score dynamics into persistent firm-specific creditworthiness, common aggregate movements, and transitory firm-specific variation. [4]

The model produces two estimates of persistent creditworthiness, which differ in the information they use and their purpose. Smoothed estimates incorporate information from a firm's entire observed history. By using both past and future observations, they provide a benchmark for evaluating how closely existing ratings reflect persistent creditworthiness. However, these estimates cannot be used to assign ratings in real time because they rely on information that was not yet available at the rating date. Filtered estimates overcome this limitation by using only information available up to that date. Although they do not benefit from future observations, they offer a feasible way to incorporate persistent creditworthiness into rating assignments.

I apply the proposed framework to a unique panel of internal credit ratings for Spanish non-financial firms compiled by the Bank of Spain, covering the period 2000–2023 and spanning firms of all sizes. These ratings are used in the Bank of Spain's lending operations with private banks to determine the eligibility and valuation of firm-level loans pledged as collateral. As a result, they are designed to respond to changes in current credit conditions, making them a natural setting in which to study the distinction between short-term and long-term creditworthiness. Furthermore, the broad coverage across firm sizes makes it possible to examine whether the importance of long-term risk differentiation differs systematically between large firms and SMEs.

The results show that current and persistent creditworthiness lead to materially different borrower classifications, particularly among smaller firms. PIT ratings differ from the model-based smoothed TTC benchmark in 51.2% of firm-year observations. These differences are less frequent among large firms (37.2%) but increase as firm size declines, reaching 52.3%

[4] Throughout the paper, long-term creditworthiness refers to persistent credit quality evaluated at average aggregate conditions, rather than a directly estimated probability of default over a particular multiyear horizon.

among micro firms. Smoothed TTC ratings also outperform PIT ratings in differentiating future defaults at every horizon and across all firm sizes, supporting their use as a benchmark for persistent creditworthiness.

The proposed method substantially mitigates these differences using only information available up to the rating date. Filtered TTC ratings reduce disagreement with the smoothed benchmark from 51.2% to 27.9% of observations. More importantly, for medium, small, and micro firms, filtered ratings increasingly outperform PIT ratings in differentiating future defaults as the horizon lengthens, although they offer similar performance at shorter horizons. In contrast, for large firms filtered ratings do not improve risk differentiation at any horizon. Thus, incorporating persistent creditworthiness into rating assignments using current and past scores can improve long-term risk differentiation, although the benefits depend on firm size.

The results point to firm-specific transitory variation as an important explanation for both the differences in classification and the gains from filtering. Transitory shocks are relatively more important in the credit-score dynamics of smaller firms, whereas large-firm scores exhibit greater persistence. Removing common year–size variation alone leaves most of the disagreement with the smoothed benchmark intact and produces only modest improvements in long-horizon risk differentiation. Consistent with these findings, the financial characteristics entering PIT scores also differ substantially in their persistence: profitability and firm size exhibit relatively short-lived movements among smaller firms, whereas the components of large-firm scores generally evolve more persistently.

Importantly, these improvements do not arise simply from producing more persistent ratings. A three-period moving average of PIT scores generates rating persistence similar to that of filtered TTC ratings, but remains further from the smoothed benchmark and performs consistently worse in differentiating future defaults. The gains from filtering therefore reflect more than the mechanical smoothing of recent credit scores.

Finally, allowing firms to differ in their exposure to GDP growth leaves the main conclusions essentially unchanged. Industry- and firm-level heterogeneity in cyclical exposure generates few additional reclassifications and virtually no improvement in risk differentiation or rating persistence. Although heterogeneous cyclicality can affect borrowers' rating assignments, it does not appear to be an important driver of the differences between PIT and TTC ratings in these data.

A large literature examines how the assessment horizon and rating-adjustment policies affect the stability and predictive accuracy of credit ratings. Altman and Rijken (2004) show

that the stability of agency ratings reflects a longer-term assessment of creditworthiness but, above all, conservative rating-migration policies, while Löffler (2013) finds that agency ratings contain information about long-run trends in market-based default risk. Kiff, Kisser, and Schumacher (2013) investigate the trade-off between rating stability and default-prediction accuracy. More recently, Gredil, Kapadia, and Lee (2022) show that agency ratings disregard transitory credit-risk shocks captured by market-based measures; although the latter predict defaults better at short horizons, ratings provide additional predictive information across horizons. These contributions establish the relevance of the assessment horizon and of the distinction between persistent and transitory credit risk. However, they focus on agency ratings, leaving open how this distinction affects borrower classifications and long-term risk differentiation in internal scoring systems.

This distinction is particularly relevant for risk-sensitive capital regulation. Banks' internal ratings have traditionally been based on borrowers' current condition (Treacy and Carey, 2000), and while the IRB framework requires probabilities of default to be calibrated to long-run default experience, this does not ensure that the underlying classifications reflect persistent creditworthiness. Existing research has examined the reliability of default-risk quantification (Carey and Hrycay, 2001), the procyclicality of regulatory capital requirements and ways of mitigating it (Gordy and Howells, 2006; Repullo, Saurina, and Trucharte, 2010), and the optimal sensitivity of capital requirements to regulatory risk assessments (Ahnert, Chapman, and Wilkins, 2021).[5] Some approaches to mitigating procyclicality preserve borrowers' relative risk rankings by construction: in Gordy and Howells (2006), for instance, TTC ratings preserve the ordering of borrowers implied by PIT assessments. Rubtsov and Petrov (2016) explicitly distinguish rating assignment from PD calibration and propose constructing TTC rating grades by adjusting the thresholds that map credit scores into ratings to offset cyclical movements. This paper examines how much borrower classifications differ when risk differentiation reflects persistent rather than current creditworthiness, and evaluates the implications of these differences for long-term default-risk differentiation.

The analysis also relates to approaches for identifying long-term creditworthiness and incorporating it into ratings. Löffler (2004) uses a permanent–transitory decomposition and the Kalman filter to study TTC ratings in a structural model of default, showing through simulations that such ratings are more stable but less effective at predicting short-term de-

[5] A related literature documents inconsistencies in banks' internal risk estimates and the incentives behind them (Behn, Haselmann, and Vig, 2022, 2025; Berg and Koziol, 2017; Mariathasan and Merrouche, 2014; Plosser and Santos, 2018).

faults than current-condition ratings. Other contributions reconcile PIT and TTC assessments through adjustments for aggregate economic conditions (Aguais, Forest, King, Lennon, and Lordkipanidze, 2006; Aguais, Forest, Wong, and Diaz-Ledezma, 2004; Fidalgo, Martínez, Moral Naranjo, and Oroz, 2024), while Cesaroni (2015) examines how smoothing estimated default probabilities affects rating migrations and stability. Building on this literature, I use a panel of internal credit scores covering firms of all sizes to quantify differences between current and persistent borrower classifications, and to construct alternative ratings based on filtered estimates of persistent creditworthiness. By separating aggregate from firm-specific transitory variation, I examine which sources of short-term credit-score movements account for these differences and whether removing them improves long-term risk differentiation across firms of different sizes.

The remainder of the paper is organized as follows. Section 2 presents an illustrative model showing why long-term creditworthiness should be incorporated during the risk differentiation stage and how transitory shocks and heterogeneous cyclical exposure can affect the long-term ranking of borrowers. Section 3 describes the institutional setting and the data. Section 4 introduces the baseline model and outlines the estimation strategy used to separate persistent firm-specific creditworthiness from transitory and cyclical components. Section 5 quantifies rating reclassifications and evaluates their implications for default-risk differentiation and persistence. Section 6 investigates the aggregate and firm-specific variation underlying the results and its association with the financial components of credit scores. Section 7 extends the framework to heterogeneous cyclical exposure. Section 8 concludes.

2 An illustrative model

This section uses a simple model to illustrate why incorporating long-term creditworthiness exclusively at the risk-quantification stage may be insufficient. When credit scores reflect both persistent creditworthiness and transitory fluctuations, classifications based on current default risk may differ from those intended to capture long-term creditworthiness.

Consider two firms, A and B. Let $y_{i,t}$ denote a transformed credit score, with higher values indicating lower current default risk. Suppose that the score can be decomposed as

$$y_{i,t} = \eta_{i,t} + \lambda_i X_t + \sigma_\varepsilon \varepsilon_{i,t}, \tag{1}$$

$$\eta_{i,t} = \eta_{i,t-1} + \sigma_\nu \nu_{i,t}, \tag{2}$$

where $\eta_{i,t}$ represents persistent firm-specific creditworthiness, X_t is an aggregate business-

cycle variable, and λ_i measures firm i's exposure to aggregate conditions. The persistent component follows a random walk, while $\varepsilon_{i,t}$ is a transitory firm-specific shock. For simplicity, assume that X_t has zero mean, so that $\eta_{i,t}$ represents the firm's creditworthiness at average aggregate conditions.

The observed scores rank firms according to their current creditworthiness, whereas a long-term classification should instead reflect their persistent components. Suppose that $\eta_{A,t} > \eta_{B,t}$, so that firm A has better persistent creditworthiness than firm B. Their ordering is reversed whenever $y_{A,t} < y_{B,t}$. Assuming that the transitory shocks are independent standard normal random variables, the probability of such a reversal is

$$\Pr(y_{A,t} < y_{B,t} \mid \eta_{A,t}, \eta_{B,t}, X_t) = \Phi\left(\frac{\eta_{B,t} - \eta_{A,t} + (\lambda_B - \lambda_A)X_t}{\sqrt{2}\sigma_\varepsilon}\right), \tag{3}$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function.

Equation 3 illustrates two sources of ranking reversals. First, transitory firm-specific shocks can overturn the persistent ordering of borrowers, particularly when their underlying creditworthiness is similar or transitory volatility is high. Second, heterogeneous exposure to aggregate conditions can change their relative ordering over the business cycle, even without corresponding changes in persistent creditworthiness. When firms have identical cyclical exposure, $\lambda_A = \lambda_B$, the aggregate component cancels from their relative ranking, as would a common time fixed effect. However, firm-specific transitory shocks can still generate ranking reversals.

Crucially, these reversals cannot be corrected simply by recalibrating the probabilities of default assigned to rating grades. Risk quantification changes the probabilities attached to existing classifications, but does not change the underlying assignment of borrowers to grades. Thus, when current credit scores rank borrowers differently from their persistent creditworthiness, adjusting the probabilities of default cannot restore their long-term ordering. Addressing this discrepancy requires incorporating persistent creditworthiness at the risk-differentiation stage itself.

This distinction motivates the two empirical exercises developed in the paper. First, I quantify how frequently PIT ratings differ from classifications based on estimated persistent creditworthiness, using smoothed estimates as a retrospective benchmark. Second, I examine whether these differences can be mitigated using filtered estimates constructed from current and past credit scores. In both cases, the original rating thresholds are maintained, isolating changes in risk differentiation from the subsequent calibration of default probabilities.

3 Data

The analysis uses a unique dataset of internal credit ratings for Spanish non-financial firms compiled by the Bank of Spain (Auria et al., 2021; Gavilá, Maldonado, and Marcelo, 2020). The data cover the period 2000–2023 and include firms ranging from large corporations to microenterprises. This broad coverage is particularly valuable for the present analysis because it allows the model to compare the relative importance of persistent and transitory creditworthiness across very different types of firms. Credit ratings are observed throughout the sample period, while information on defaults and insolvency proceedings is available from 2004 onward.

Since identification of the persistent component requires observing firms more than once, firms appearing in only one year are excluded. The final sample contains 19,136,969 firm-year observations corresponding to 1,992,716 unique firms, of which 17,487,037 include default information.

Each firm-year observation contains the internal credit rating assigned by the Bank of Spain together with the underlying continuous credit score on which that rating is based. In addition, the data include the firm's size classification,[6] economic sector, indicators for default and insolvency proceedings, an indicator for whether insolvency proceedings begin in the following year, and the share of credit exposure reported as defaulted in the following year according to the Central Credit Register (CCR). The data also contain the contributions of different financial characteristics to the credit score, which sum to the overall score, although the underlying financial ratios and other balance-sheet variables are not themselves included in the dataset.

For the analysis of rating transitions, I augment the original rating scale with two states. The first corresponds to default (D), assigned whenever the firm defaults during the current year. The second corresponds to bankruptcy (Q), assigned to the final observation of a firm that exits the dataset after either entering insolvency proceedings or defaulting during the current or subsequent year. Note that the latter is an absorbing state. These states replace the reported rating whenever the corresponding event occurs and allow transition matrices to account for exits associated with financial distress.

Credit scores are generated from logistic models estimated on firms' balance-sheet infor-

[6] Firm size is defined according to Commission Recommendation 2003/361/EC (Annex, Art. 2). Micro firms have fewer than 10 employees and annual turnover and/or total assets not exceeding €2 million; small firms have fewer than 50 employees and turnover and/or total assets not exceeding €10 million; and medium-sized firms are the remaining SMEs, with fewer than 250 employees and turnover not exceeding €50 million and/or total assets not exceeding €43 million. Firms that do not satisfy the SME criteria are classified as large.

mation. Because the determinants of credit risk differ substantially across firms, the Bank of Spain estimates separate scoring models for large firms, medium and small firms, and micro firms. Within each model, the resulting scores are normalized to lie between 0 and 100, with higher values indicating better credit quality, and are subsequently mapped into ordered rating categories using historical default rates.

This institutional feature plays an important role in the empirical model. As firms grow or shrink over time, they may move between scoring models, creating mechanical changes in their reported scores that do not necessarily reflect changes in underlying creditworthiness. Section 4 explicitly accounts for these model transitions through a set of correction parameters.

Although the logistic model does not explicitly include macroeconomic variables, it may still capture aggregate conditions implicitly through the use of firm-level balance sheet information, which itself may reflect the business cycle. The underlying financial data are drawn from two main sources: the Central Balance Sheet Data Office (CBI), an internal Bank of Spain database containing financial statements filed with the Mercantile Registry, and INFORMA, a private provider of financial and payment information on Spanish firms. Ratings are assigned to all non-financial firms with valid balance sheet data, regardless of firm size or public listing status. Data validity is assessed according to internal Eurosystem rules; for example, financial statements cannot be more than two years old at the time of rating assignment.

Sample	Order	4+	4	4-	5+	5	5-	6+	6	6-	7+	7	7-	D
All firms	1st order	0.46	0.73	0.55	0.61	0.31	0.20	0.43	0.18	0.27	0.29	0.40	0.16	0.67
	2nd order	0.35	0.68	0.44	0.57	0.27	0.16	0.39	0.15	0.23	0.24	0.34	0.10	0.51
Large	1st order	0.46	0.61	0.55	0.47	0.38	0.49	0.27	0.19	0.23	0.37	0.22	0.12	0.71
	2nd order	0.35	0.54	0.44	0.37	0.29	0.39	0.20	0.14	0.14	0.26	0.14	0.05	0.57
Medium	1st order	—	0.76	—	0.39	—	0.21	0.25	0.21	—	0.24	0.07	0.23	0.73
	2nd order	—	0.72	—	0.34	—	0.17	0.22	0.17	—	0.19	0.06	0.17	0.58
Small	1st order	—	0.73	—	0.35	—	0.18	0.25	0.22	—	0.26	0.09	0.22	0.70
	2nd order	—	0.68	—	0.31	—	0.14	0.23	0.18	—	0.21	0.09	0.14	0.52
Micro	1st order	—	—	—	0.66	0.31	—	0.45	0.17	0.27	0.29	0.42	0.10	0.69
	2nd order	—	—	—	0.62	0.27	—	0.40	0.15	0.23	0.25	0.36	0.06	0.50

Table 1: Diagonal elements of the first- and second-order transition matrices for PIT ratings. Element i is estimated as the number of transitions to the same category i divided by the total number of transitions starting in i.

Table 1 reports the diagonal elements of the estimated first- and second-order transition matrices. These are constructed under a discrete multinomial framework: transition probabilities are computed as the number of observed transitions from rating category i to category

j divided by the total number of transitions originating from category i. The first-order transition matrix captures one-year rating dynamics, while the second-order matrix reflects transitions over a two-year horizon.

Consistent with the credit ratings literature, diagonal elements dominate the transition matrices, although they are not particularly high. This indicates that ratings respond relatively strongly to changes in firms' conditions, a feature characteristic of point-in-time (PIT) ratings. This behavior is expected given that these ratings are primarily used to assess asset eligibility for monetary policy operations. The second-order transition matrix exhibits limited decay in the diagonal elements, indicating some persistence in rating dynamics. This persistence can be exploited to extract an underlying through-the-cycle (TTC) component of creditworthiness that smooths both transitory idiosyncratic shocks and business-cycle fluctuations.

The availability of both continuous credit scores and repeated observations for the same firms makes it possible to distinguish persistent movements in creditworthiness from transitory fluctuations. The next section develops a state-space model that exploits this panel structure while accounting for changes in the underlying scoring models as firms move across size categories.

4 Methodology

Let $i \in \{1, \ldots, n\}$ index firms and $t \in \{1, \ldots, T\}$ index time periods. Firm i is observed over a subset of periods $T_i \subseteq \{1, \ldots, T\}$. Let $j_{it} \in \mathcal{G} := \{4+, 4, \ldots, 7-\}$ denote the observed ordered credit rating of firm i at time t, $s_{it} \in \mathcal{S} := \{\text{Large, Medium, Small, Micro}\}$ its size category, and $m_{it} \in \mathcal{M} := \{\text{Large, Medium/Small, Micro}\}$ the scoring model used to compute its credit score.[7]

I model the evolution of firms' credit scores as the sum of a common time–size component and firm-specific dynamics.[8] Given a credit score p_{it} normalized to lie between 0 and 100, I

[7] The model conditions on the periods in which a firm is observed and on its observed history of size classifications. This assumption is standard in the credit ratings literature, where no information is inferred from unobserved periods.

[8] For a version of this framework applicable when only discrete credit ratings are observed, rather than the underlying continuous scores, see Peña (2025). That formulation uses an ordered-probit model and is estimated using stochastic EM and MCMC methods.

first apply a logit transformation [9] and decompose it as:

$$y_{it} = \log\left(\frac{p_{it}}{100 - p_{it}}\right), \tag{4}$$

$$y_{it} = \eta_{it} + \delta_{t,s_{it}} + \xi_{it}, \tag{5}$$

where $\delta_{t,s_{it}}$ is a time–size-specific fixed effect capturing aggregate conditions affecting firms within the same size category, η_{it} is a persistent firm-specific component, and ξ_{it} is a transitory idiosyncratic component. The specification is therefore linear in the logit-transformed scores, but nonlinear in the original credit scores.

The persistent component follows a random walk, while the transitory component consists of a white-noise shock and a correction term accounting for changes in the scoring model:

$$\eta_{it} = \eta_{i,t-1} + \nu_{it}, \tag{6}$$

$$\xi_{it} = \varepsilon_{it} + \beta_{it}, \tag{7}$$

$$\beta_{it} = \begin{cases} 0, & \text{if } m_{it} = m_{i,t-1}, \\ \beta_{m_{it},j_{it},m_{i,t-1},j_{i,t-1}}, & \text{if } m_{it} \neq m_{i,t-1}. \end{cases} \tag{8}$$

The correction term β_{it} captures discrete changes in scores arising from switches between scoring models and their associated rating thresholds.[10] The stochastic components η_{i0}, ε_{it} and ν_{it} are assumed to be mutually independent across firms and time, and normally distributed with size-specific variances $\sigma^2_{0,s_{i0}}$, $\sigma^2_{\varepsilon,s_{it}}$ and $\sigma^2_{\eta,s_{it}}$, respectively.

The model is estimated by maximum likelihood using a state-space representation together with the Kalman smoother within an Expectation–Maximization (EM) algorithm (see Shumway and Stoffer (1982)). Given the estimated parameters, the Kalman filter and smoother recover the conditional expectation of the persistent component η_{it} using either information available up to period t (filtered estimate) or the full sample (smoothed estimate). I define the through-the-cycle (TTC) log-odds measure as

$$y^{TTC}_{it} = \mathbb{E}[\eta_{it} \mid \mathcal{I}_\tau] + \bar{\delta}_{s_{it}} + \beta_{it}, \tag{9}$$

where $\mathcal{I}_\tau$ denotes the information set available up to $\tau \in \{t, T\}$ and $\bar{\delta}_{s_{it}}$ is the average fixed effect for firms in size category s_{it}. TTC credit scores are then obtained by applying the inverse log-odds transformation, while TTC ratings are obtained by applying the original

[9] Scores equal to 0 and 100 are adjusted by adding or subtracting a small constant before applying the transformation. This affects approximately 1% of the observations.

[10] For example, when a firm changes size category (e.g. from medium to large), both the scoring model and the score scale change. As a result, the numerical score may shift substantially even if the firm's underlying credit quality, as reflected by its rating, remains unchanged. The correction term, β_{it}, accounts for these mechanical discontinuities.

rating thresholds to the resulting TTC scores.

Using the original thresholds helps isolate the role of risk differentiation. The rating scale remains unchanged, but firms may be assigned to different grades because scores are now based on long-term rather than current creditworthiness. These differences therefore capture adjustments occurring at the risk-differentiation stage, which cannot be replicated by merely changing the probability of default attached to each existing grade.

The filtered and smoothed estimates serve distinct purposes. Conditional on the estimated model parameters, smoothed estimates use a firm's entire observed credit-score history, including observations after the rating date. They therefore provide the most informative retrospective estimate of persistent creditworthiness under the maintained model, but cannot be used to assign ratings in real time. I use the resulting smoothed TTC ratings as a benchmark to quantify how much various PIT classifications differ from those based on persistent creditworthiness. Filtered estimates, by contrast, use only information available up to the rating date. Although they do not benefit from future observations, they provide a feasible way to incorporate persistent creditworthiness into rating assignments. I therefore evaluate filtered TTC ratings as an alternative to PIT ratings, examining both their agreement with the retrospective benchmark and their ability to differentiate future defaults.

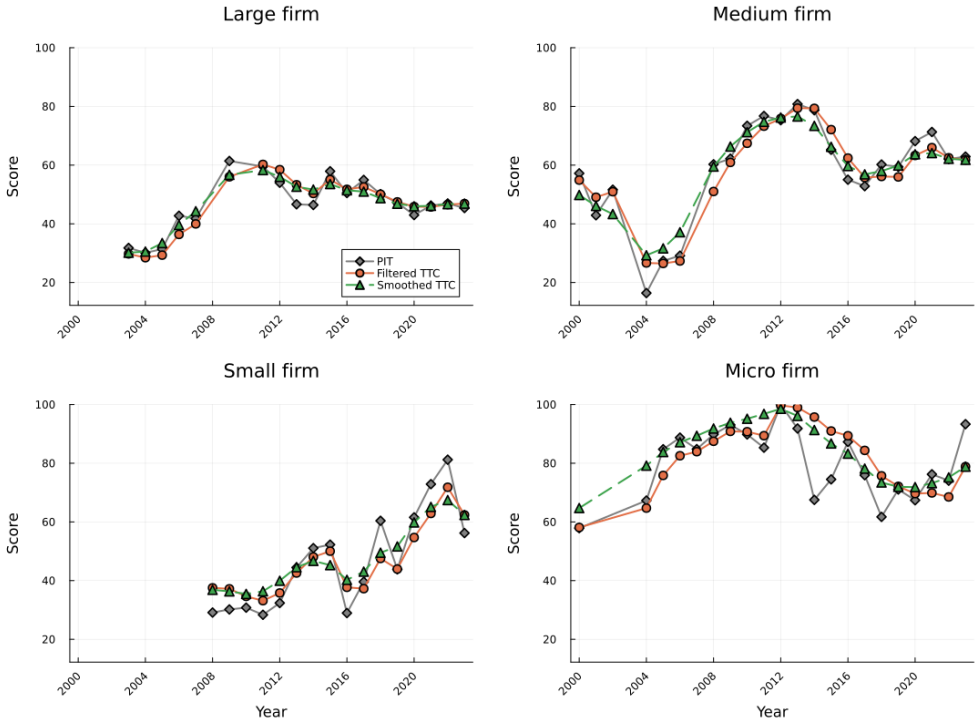


Figure 1: Evolution of PIT, filtered TTC and smoothed TTC credit scores for four firms randomly selected among those observed for more than 10 periods and with constant size.

Figure 1 illustrates the behaviour of the estimated TTC scores for four firms randomly selected among those observed for more than 10 periods and with constant size. Relative to the original PIT scores, both the filtered and smoothed TTC scores exhibit substantially less short-run variation. This reflects the objective of the model: to remove fluctuations associated with aggregate business-cycle conditions and transitory firm-specific shocks while preserving the persistent component of firm creditworthiness.

5 Long-Term Rating Reclassifications and Risk Differentiation

Section 2 explained why classifications based on current creditworthiness may be inappropriate for a long-term rating objective. This section assesses the empirical importance of this distinction and whether it can be addressed using information available at the rating date. I first quantify how frequently PIT ratings differ from classifications based on the smoothed TTC benchmark and examine whether the latter better differentiate future default risk. I then evaluate whether filtered TTC ratings can mitigate these differences and improve risk differentiation using only current and past credit scores.

Sample	Pair	All			> 10		
		N	Diff.	Share	N	Diff.	Share
All firms	PIT vs TTC s.	19136969	9802386	51.2%	11858980	5723673	48.3%
	MA 3 vs TTC s.	19136969	8594934	44.9%	11858980	5049008	42.6%
	TTC f. vs TTC s.	19136969	5331934	27.9%	11858980	3451629	29.1%
Large	PIT vs TTC s.	79860	29692	37.2%	63335	23368	36.9%
	MA 3 vs TTC s.	79860	32226	40.4%	63335	25625	40.5%
	TTC f. vs TTC s.	79860	17724	22.2%	63335	15235	24.1%
Medium	PIT vs TTC s.	371124	164521	44.3%	281978	122509	43.4%
	MA 3 vs TTC s.	371124	149837	40.4%	281978	112713	40.0%
	TTC f. vs TTC s.	371124	97447	26.3%	281978	77617	27.5%
Small	PIT vs TTC s.	2378761	1071968	45.1%	1760117	773577	44.0%
	MA 3 vs TTC s.	2378761	1020969	42.9%	1760117	733903	41.7%
	TTC f. vs TTC s.	2378761	630843	26.5%	1760117	481510	27.4%
Micro	PIT vs TTC s.	16307224	8536205	52.3%	9753550	4804219	49.3%
	MA 3 vs TTC s.	16307224	7391902	45.3%	9753550	4176767	42.8%
	TTC f. vs TTC s.	16307224	4585920	28.1%	9753550	2877267	29.5%

Table 2: The table reports the number and share of firm-year observations assigned to different rating grades by each pair of rating systems. Results are presented for the full sample and for a subsample of firms observed for more than ten periods, excluding each firm's final observation.

Table 2 quantifies the extent of reclassification, using smoothed TTC ratings as the benchmark for persistent creditworthiness. As a simple alternative, I also construct an equally weighted three-period moving average (MA 3) of the PIT score, using the current and up to

two preceding observations for each firm, when available. This allows me to assess whether the differences between PIT and TTC classifications can be reduced by mechanically smoothing recent scores.

PIT ratings differ from the smoothed TTC classification in 51.2% of observations. Disagreement is smaller among large firms (37.2%) and increases as firm size declines, reaching 52.3% among micro firms. Filtered TTC ratings substantially reduce these differences: their disagreement with the smoothed benchmark is 27.9% overall, compared with 44.9% for the MA 3 rating. Thus, filtering brings classifications considerably closer to the retrospective benchmark than simply averaging recent PIT scores.

The right-hand columns restrict the sample to firms observed for more than ten periods, for which the smoothed estimate is based on a longer firm history, and exclude the final observation, where filtered and smoothed estimates coincide mechanically. The results are similar: PIT ratings differ from the smoothed benchmark in 48.3% of observations, compared with 29.1% for filtered TTC ratings. The differences therefore do not appear to be driven by firms with short histories or by the mechanical convergence of filtered and smoothed estimates at the end of the sample.

These results indicate that current and persistent creditworthiness imply substantially different borrower classifications. However, reclassification alone does not establish whether the estimated persistent component provides an economically meaningful benchmark. To assess this, table 3 compares the ability of smoothed TTC and PIT ratings to differentiate future default risk using the area under the receiver operating characteristic curve (AUROC),[11] for defaults within one, three, six, and ten years.[12] The table reports differences in AUROC relative to PIT ratings, with positive values indicating better risk differentiation.

Smoothed TTC ratings outperform PIT ratings at every horizon and across all firm sizes. Moreover, the improvements generally increase with the horizon. For the full sample, the gain rises from 1.2 percentage points at the one-year horizon to 4.8 percentage points at the ten-year horizon. The improvements are particularly pronounced among medium, small, and

[11] The AUROC measures a rating system's ability to discriminate between firms that default and those that do not. It can be interpreted as the probability that a randomly selected defaulting firm receives a worse rating than a randomly selected non-defaulting firm, with ties counted as one-half. An AUROC of 0.5 indicates no discriminatory power, while a value of 1 indicates perfect discrimination.

[12] A firm-year (i, t) enters the h-year sample if the whole window $t+1, \ldots, t+h$ lies within the period covered by the default data (e.g. rating years up to 2014 for $h = 10$). It is classified as a default if the firm is in default in the Bank of Spain's Central Credit Register (CIR) at any point during the window. Because the CIR records defaults whether or not the firm is rated, this includes firms that stop being rated before or when they default. It is classified as a non-default if no default is recorded during the window and the firm is still observed in the data at the end of the window, i.e. in year $t + h - 1$ or later. Firm-years in which the firm is already in default on 31 December of the rating year are excluded.

micro firms, reaching approximately 5 percentage points at the longest horizon. By contrast, the gains are considerably smaller for large firms, reaching 1.9 percentage points after ten years.

These results support the use of smoothed TTC ratings as a retrospective benchmark for persistent creditworthiness. In particular, the larger improvements at longer horizons are consistent with the interpretation that removing transitory variation produces classifications more informative about long-term default risk. Importantly, however, smoothed ratings exploit information that was unavailable at the rating date. Their superior risk differentiation should therefore be interpreted as evidence about the retrospective benchmark, rather than as the performance of a feasible rating procedure.

	Within 1 year	Within 3 years	Within 6 years	Within 10 years
All firms	+1.2	+3.3	+4.2	+4.8
Large	+0.5	+1.4	+1.7	+1.9
Medium	+1.5	+3.4	+4.1	+4.9
Small	+1.5	+3.4	+4.3	+5.1
Micro	+1.3	+3.3	+4.2	+4.8

Table 3: The table reports the difference in AUROC between smoothed TTC and PIT ratings, in percentage points, for defaults occurring within one, three, six, and ten years of the rating date. Positive values indicate better risk differentiation by smoothed TTC ratings.

Therefore, the next question is whether persistent creditworthiness can be incorporated into rating assignments using only information available at the rating date. Figure 2 compares the risk differentiation of filtered TTC and MA 3 ratings against the original PIT ratings across default horizons. Both alternatives use only current and past credit scores, but differ in how they account for transitory variation.

Figure 2 shows that the benefits of filtering depend on both firm size and the default horizon. For medium, small, and micro firms, filtered TTC ratings provide little improvement over PIT ratings at short horizons, but increasingly outperform them as the horizon lengthens. At the ten-year horizon, the gains reach 0.9 percentage points for medium firms, 1.6 percentage points for small firms, and 0.7 percentage points for micro firms. Large firms exhibit a different pattern: filtered TTC ratings do not improve risk differentiation at any horizon. Thus, while filtering can improve long-term risk differentiation using only current and past scores, the benefits are not uniform across firms.

The MA 3 benchmark shows that these improvements are not simply a consequence of smoothing recent scores. Mechanical averaging generally weakens risk differentiation at short

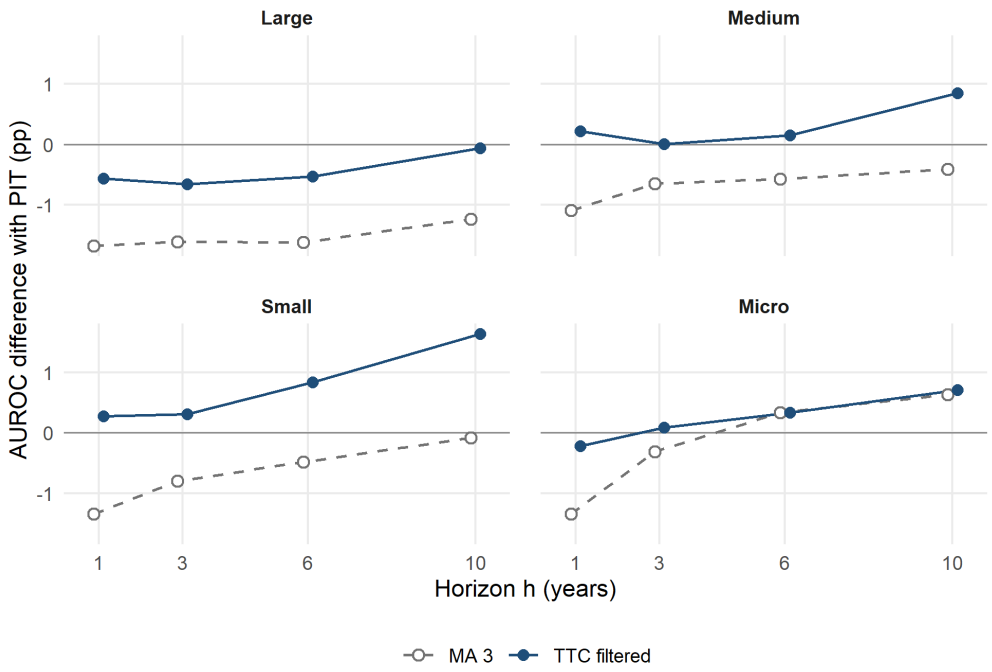


Figure 2: The figure reports differences in AUROC relative to PIT ratings, in percentage points, for defaults occurring within one, three, six, and ten years of the rating date. Positive values indicate improvements in risk differentiation relative to PIT ratings.

horizons and produces, at most, modest gains at longer horizons. It also performs worse than filtered TTC ratings across nearly all comparisons. The benefits of filtering therefore reflect more than greater rating stability: they depend on distinguishing persistent creditworthiness from transitory movements in credit scores.

Table A.1 in the Appendix A reports the complete AUROC results, including comparisons restricted to observations where PIT ratings differ from smoothed or filtered TTC ratings. The improvements obtained with smoothed TTC ratings are consistently larger in the subsample where they differ from PIT ratings. Similarly, the gains from filtered TTC ratings at longer horizons are generally more pronounced among observations whose classifications change, particularly for smaller firms. These results reinforce the economic relevance of the reclassifications: differences between current and persistent creditworthiness are associated with meaningful differences in the ability to distinguish future default risk, and the benefits of filtering can be particularly important among borrowers whose ratings would change.

Overall, TTC ratings improve the differentiation of long-term credit risk for medium, small, and micro firms, but offer limited gains for large firms. I next examine whether this greater differentiation of long-term credit risk is accompanied by the greater stability expected of a long-term rating system.

Sample	Ratings	4+	4	4-	5+	5	5-	6+	6	6-	7+	7	7-	D
All firms	PIT	0.46	0.73	0.55	0.61	0.31	0.20	0.43	0.18	0.27	0.29	0.40	0.16	0.67
	MA 3	0.61	0.85	0.65	0.78	0.55	0.35	0.67	0.33	0.44	0.46	0.59	0.28	0.67
	TTC f.	0.60	0.84	0.66	0.78	0.58	0.33	0.69	0.37	0.50	0.50	0.71	0.33	0.67
	TTC s.	0.71	0.89	0.74	0.87	0.74	0.46	0.81	0.60	0.71	0.69	0.83	0.44	0.67
Large	PIT	0.46	0.61	0.55	0.47	0.38	0.49	0.27	0.19	0.23	0.37	0.22	0.12	0.71
	MA 3	0.61	0.73	0.65	0.57	0.51	0.65	0.43	0.29	0.38	0.56	0.42	0.36	0.71
	TTC f.	0.60	0.73	0.66	0.58	0.49	0.61	0.37	0.24	0.31	0.49	0.39	0.40	0.71
	TTC s.	0.71	0.81	0.74	0.67	0.58	0.70	0.47	0.32	0.41	0.57	0.43	0.33	0.71
Medium	PIT	—	0.76	—	0.39	—	0.21	0.25	0.21	—	0.24	0.07	0.23	0.73
	MA 3	—	0.89	—	0.61	—	0.37	0.44	0.39	—	0.46	0.17	0.40	0.73
	TTC f.	—	0.86	—	0.59	—	0.35	0.42	0.37	—	0.44	0.18	0.39	0.73
	TTC s.	—	0.92	—	0.73	—	0.51	0.59	0.54	—	0.62	0.31	0.50	0.73
Small	PIT	—	0.73	—	0.35	—	0.18	0.25	0.22	—	0.26	0.09	0.22	0.70
	MA 3	—	0.84	—	0.55	—	0.32	0.44	0.41	—	0.50	0.22	0.37	0.70
	TTC f.	—	0.83	—	0.54	—	0.30	0.41	0.36	—	0.44	0.18	0.31	0.70
	TTC s.	—	0.89	—	0.68	—	0.44	0.56	0.51	—	0.59	0.31	0.43	0.70
Micro	PIT	—	—	—	0.66	0.31	—	0.45	0.17	0.27	0.29	0.42	0.10	0.69
	MA 3	—	—	—	0.84	0.55	—	0.68	0.32	0.44	0.45	0.60	0.18	0.69
	TTC f.	—	—	—	0.84	0.58	—	0.70	0.37	0.50	0.52	0.75	0.34	0.69
	TTC s.	—	—	—	0.92	0.74	—	0.83	0.61	0.71	0.72	0.87	0.47	0.69

Table 4: Diagonal elements of the first-order transition matrices for PIT, MA 3, filtered and smoothed TTC ratings. Element i is estimated as the number of transitions to the same category i divided by the total number of transitions starting in i.

Table 4 reports one-period transition matrices for PIT, MA 3, filtered TTC, and smoothed TTC ratings. Both TTC measures are substantially more persistent than PIT ratings, as shown by the larger diagonal elements of their transition matrices. The increase is particularly pronounced for the smoothed TTC rating, which uses the full firm history to estimate persistent creditworthiness. Filtered TTC ratings also display substantially greater persistence.

The MA 3 ratings also display substantially greater persistence than PIT ratings. Their diagonal transition probabilities are close to those of filtered TTC ratings, and for large, medium, and small firms they are larger in most rating categories. Only among micro firms are filtered TTC ratings clearly more persistent. A substantial increase in rating persistence can therefore be obtained through simple mechanical smoothing.

However, as shown in Figure 2, this greater persistence does not translate into better risk differentiation: MA 3 ratings differentiate risk worse than filtered TTC ratings for every size class and at every horizon. Small firms illustrate this most clearly. Their MA 3 ratings are more persistent than their filtered TTC ratings in every rating category, yet they differentiate risk worse at every horizon. Improving long-term risk differentiation therefore depends less on how persistent the ratings are than on which variation is removed from the credit score.

To quantify these differences more formally, I use the persistence measure proposed by Jafry and Schuermann (2004). Let P denote a transition matrix and define the mobility matrix as $\tilde{P} := P - I$, where I is the identity matrix. The mobility measure is given by the average singular value of the mobility matrix:

$$M_{\text{SVD}}(P) := \frac{1}{K} \sum_{k=1}^{K} \sqrt{\lambda_k\left(\tilde{P}'\tilde{P}\right)}, \tag{10}$$

where $\lambda_k(A)$ denotes the kth eigenvalue of matrix A and K is the number of rating categories. Lower values of $M_{\text{SVD}}(P)$ correspond to more persistent rating dynamics.

Order	Ratings	All firms	Large	Medium	Small	Micro
1st order	PIT	0.582	0.586	0.437	0.442	0.421
	MA 3	0.451	0.461	0.346	0.348	0.331
	TTC f.	0.418	0.477	0.350	0.368	0.279
	TTC s.	0.296	0.415	0.266	0.289	0.185
2nd order	PIT	0.641	0.672	0.472	0.479	0.457
	MA 3	0.588	0.611	0.435	0.441	0.422
	TTC f.	0.537	0.596	0.422	0.440	0.369
	TTC s.	0.442	0.563	0.372	0.394	0.290

Table 5: Persistence measure proposed by Jafry and Schuermann (2004) for 1st and 2nd order transition matrices, for all firms and for each size category. Lower values are associated with greater persistence in the transition matrix.

Table 5 reports the mobility measure for first- and second-order transition matrices. The results confirm that both TTC ratings are more persistent than PIT ratings, with the smoothed TTC rating displaying the lowest mobility for every firm-size category. MA 3 and filtered TTC occupy an intermediate position. Their persistence is similar overall, and neither uniformly dominates across size groups.

Parameter	Large	Medium	Small	Micro
$\sigma_{0,s}$	0.74	0.56	0.57	0.51
$\sigma_{\eta,s}$	0.35	0.29	0.32	0.42
$\sigma_{\varepsilon,s}$	0.39	0.41	0.43	0.86
$\sigma_{\eta,s}/\sigma_{\varepsilon,s}$	0.91	0.72	0.75	0.49

Table 6: Estimated standard deviation parameters by firm size.

The estimated variance components help explain why the differences between PIT and TTC ratings, and the benefits of filtering, vary across firm sizes. Table 6 reports the standard deviations of innovations to persistent creditworthiness and of transitory shocks. Transitory shocks are relatively less important for large firms, where the ratio $\sigma_{\eta,s}/\sigma_{\varepsilon,s}$ is 0.91, compared with 0.72 for medium firms, 0.75 for small firms, and 0.49 for micro firms. This pattern is

consistent with the lower reclassification rates observed among large firms and the smaller improvements in risk differentiation obtained with both smoothed and filtered TTC ratings. When transitory fluctuations are relatively less important, current credit scores provide a more informative signal of persistent creditworthiness, leaving less scope for improving long-term risk differentiation by removing these fluctuations.

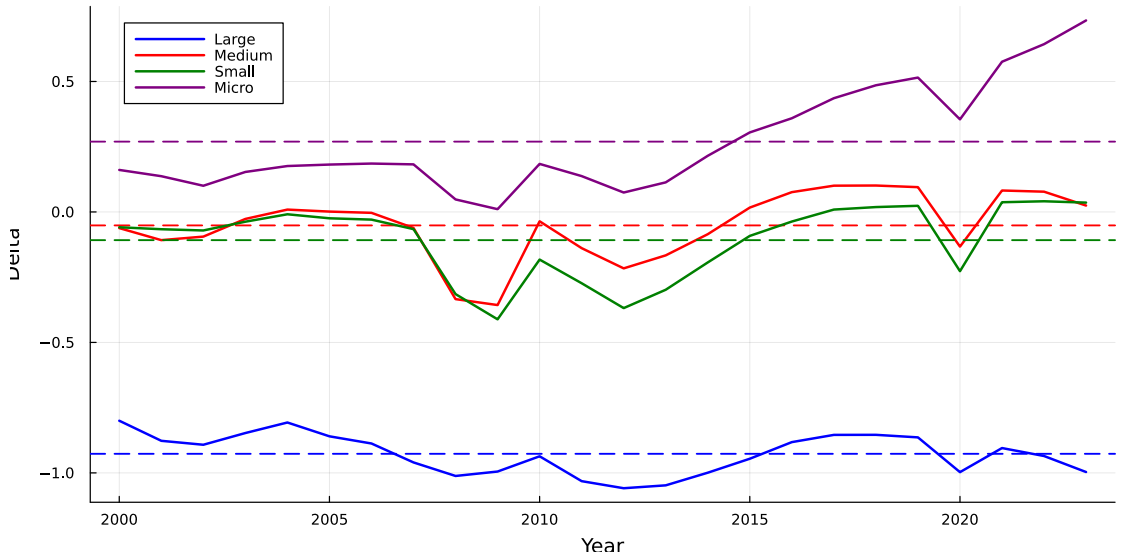


Figure 3: The figure displays the estimated year–size fixed effects, $(\delta_{t,s})$, over 2000–2023. Dashed horizontal lines indicate the corresponding averages over the sample period. Higher values reflect improvements in average credit quality within a given firm-size category.

Figure 3 presents the estimated year–size fixed effects, where lower values correspond to a deterioration in average credit quality. The estimates display broadly procyclical dynamics across firm sizes, with marked deteriorations around the Global Financial Crisis, the European sovereign debt crisis, and the COVID-19 shock. These common cyclical movements are particularly apparent for large, medium, and small firms, whereas micro firms exhibit a more pronounced long-run improvement in credit quality during the second half of the sample.

Credit quality is relatively stable during the first half of the 2000s, before deteriorating sharply around the Global Financial Crisis. The decline is especially pronounced for medium and small firms in 2008–09, while the adjustment is earlier and more gradual for large firms. After a temporary recovery around 2010, credit quality deteriorates again during the European sovereign debt crisis, reaching another trough around 2012–13. This is followed by a sustained recovery through the second half of the 2010s. The COVID-19 shock produces a visible but short-lived deterioration in 2020 across all firm sizes, followed by a rapid rebound in 2021.

Micro firms exhibit a noticeably different longer-run pattern. Their estimated year effects also deteriorate during the Global Financial Crisis and the sovereign debt crisis, but from

around 2013 onward they increase persistently and substantially, reaching levels well above their historical average by the end of the sample. By contrast, the recovery among large, medium, and small firms is more limited, and their estimated effects remain comparatively stable after the post-sovereign-debt-crisis rebound. The resulting divergence suggests that the evolution of average credit quality over the later part of the sample differs substantially between micro firms and the other size categories.

The sustained improvement among micro firms may reflect structural changes in their underlying creditworthiness rather than solely cyclical fluctuations. However, the model does not distinguish persistent changes in average creditworthiness from business-cycle movements, as both are captured by the year–size fixed effects. Extending the framework to separate these sources of variation, and investigating the factors underlying the apparent improvement in micro firms' creditworthiness, are important directions for future research.

6 Sources of Short-Term Variation in Credit Scores

The previous section showed that incorporating persistent creditworthiness into rating assignments can improve long-term risk differentiation, but that the gains from filtering differ substantially across firm sizes. In particular, filtered TTC ratings increasingly outperform PIT ratings at longer horizons for medium, small, and micro firms, while offering no improvement for large firms. This section investigates the sources of credit-score variation underlying these differences. I first examine whether removing common aggregate fluctuations alone can reproduce the gains from filtering, or whether firm-specific transitory movements play a more important role. I then investigate the persistence of the financial characteristics entering PIT scores to better understand why filtering is more beneficial for smaller firms.

6.1 Aggregate versus firm-specific transitory variation

The baseline model identifies two sources of short-term variation in PIT creditworthiness: aggregate fluctuations captured by the year–size effects, $\delta_{t,s_{it}}$, and firm-specific transitory movements, ε_{it}. To assess their relative importance, I construct an intermediate rating measure that removes the former while leaving the latter unchanged. Specifically, I define the aggregate-adjusted PIT log-odds as

$$y_{it}^{adj} = y_{it} - (\delta_{t,s_{it}} - \overline{\delta}_{s_{it}}).$$

The corresponding score is obtained using the inverse log-odds transformation and mapped into rating categories using the original thresholds. This measure therefore evaluates the aggregate component at its average level while preserving all firm-specific variation in the original PIT score.

Sample	Pair	All			> 10		
		N	Diff.	Share	N	Diff.	Share
All firms	PIT vs PIT adj.	19136969	5354160	28.0%	11858980	2994356	25.2%
	TTC f. vs PIT adj.	19136969	8807678	46.0%	11858980	5102845	43.0%
	TTC s. vs PIT adj.	19136969	9265648	48.4%	11858980	5397180	45.5%
Large	PIT vs PIT adj.	79860	12465	15.6%	63335	9912	15.7%
	TTC f. vs PIT adj.	79860	24721	31.0%	63335	19253	30.4%
	TTC s. vs PIT adj.	79860	27879	34.9%	63335	21897	34.6%
Medium	PIT vs PIT adj.	371124	89406	24.1%	281978	69802	24.8%
	TTC f. vs PIT adj.	371124	136892	36.9%	281978	101071	35.8%
	TTC s. vs PIT adj.	371124	155050	41.8%	281978	114315	40.5%
Small	PIT vs PIT adj.	2378761	653678	27.5%	1760117	482898	27.4%
	TTC f. vs PIT adj.	2378761	875032	36.8%	1760117	627012	35.6%
	TTC s. vs PIT adj.	2378761	999965	42.0%	1760117	715582	40.7%
Micro	PIT vs PIT adj.	16307224	4598611	28.2%	9753550	2431744	24.9%
	TTC f. vs PIT adj.	16307224	7771033	47.7%	9753550	4355509	44.7%
	TTC s. vs PIT adj.	16307224	8082754	49.6%	9753550	4545386	46.6%

Table 7: The table reports the number and share of firm-year observations assigned to different rating grades when comparing aggregate-adjusted PIT ratings with PIT, filtered TTC, and smoothed TTC ratings. Results are presented for the full sample and for firms observed for more than ten periods, excluding each firm's final observation.

Table 7 compares the classifications implied by the original PIT, aggregate-adjusted PIT, and TTC ratings. Removing aggregate fluctuations alone changes 28.0% of PIT classifications. However, the aggregate-adjusted PIT rating remains different from the smoothed TTC benchmark in 48.4% of observations, only moderately below the 51.2% disagreement between the original PIT and smoothed TTC ratings reported in Table 2. The same pattern holds across firm-size groups: aggregate adjustment produces fewer reclassifications among large firms, but leaves substantial disagreement with the long-term benchmark at every size.

These comparisons indicate that removing common aggregate fluctuations accounts for only part of the difference between current and long-term classifications. Most of the disagreement with the smoothed benchmark remains after the aggregate component has been removed. This result points to firm-specific transitory variation as an important source of PIT–TTC differences.[13]

[13]Because rating assignments are nonlinear functions of the underlying scores, the reclassification shares should not be interpreted as an additive decomposition of the PIT–TTC gap; rather, the exercise shows how much disagreement remains when aggregate fluctuations are removed in isolation.

Table 8 examines whether removing common aggregate fluctuations alone reproduces the improvements in risk differentiation obtained with the TTC ratings. For medium, small, and micro firms, it generally does not. Aggregate adjustment has little effect on AUROC at short horizons and produces only modest gains as the horizon lengthens, considerably smaller than those obtained with the filtered TTC ratings. The more detailed results reported in Appendix A Table A.2 reinforce this pattern. Among observations for which PIT and TTC ratings disagree, aggregate adjustment improves long-horizon discrimination, but the gains remain considerably smaller than those from removing firm-specific transitory variation.

Large firms display a different pattern. Aggregate adjustment does not systematically improve on the original PIT rating, but it generally performs better than the filtered TTC rating. The smoothed TTC rating nevertheless remains the strongest long-horizon discriminator.

Taken together, the results suggest that the composition of short-term variation differs importantly across firm sizes. For medium, small, and micro firms, firm-specific transitory movements appear to be more important than common cyclical fluctuations for long-term risk differentiation. For large firms, by contrast, common aggregate conditions appear relatively more important than idiosyncratic variation, but neither source of transitory movement appears to offer much scope for improving risk differentiation when removed.

Sample	Within 1 year				Within 3 years			
	PIT	PIT adj.	TTC f.	TTC s.	PIT	PIT adj.	TTC f.	TTC s.
All firms	0.764	-0.003	-0.003	+0.012	0.727	-0.005	0.000	+0.033
Large	0.706	-0.002	-0.006	+0.005	0.690	-0.002	-0.007	+0.014
Medium	0.809	+0.001	+0.002	+0.015	0.787	-0.003	0.000	+0.034
Small	0.831	0.000	+0.003	+0.015	0.800	-0.003	+0.003	+0.034
Micro	0.753	-0.002	-0.002	+0.013	0.716	-0.003	+0.001	+0.033
Sample	Within 6 years				Within 10 years			
	PIT	PIT adj.	TTC f.	TTC s.	PIT	PIT adj.	TTC f.	TTC s.
All firms	0.705	-0.003	+0.004	+0.042	0.693	+0.001	+0.009	+0.048
Large	0.660	-0.001	-0.005	+0.017	0.630	+0.002	-0.001	+0.019
Medium	0.761	-0.002	+0.001	+0.041	0.730	+0.004	+0.008	+0.049
Small	0.773	+0.001	+0.008	+0.043	0.747	+0.008	+0.016	+0.051
Micro	0.698	-0.002	+0.003	+0.042	0.692	0.000	+0.007	+0.048

Table 8: The table reports AUROC values for PIT ratings and differences in AUROC relative to PIT ratings for aggregate-adjusted PIT, filtered TTC, and smoothed TTC ratings. Results are presented for defaults occurring within one, three, six, and ten years of the rating date. Positive differences indicate improvements in risk differentiation relative to PIT ratings.

6.2 Financial characteristics underlying transitory variation

To better understand the differences in the gains from filtering across firm sizes, I examine the persistence of the financial characteristics entering PIT scores. The construction of the PIT score makes this exercise possible. Each financial ratio is first discretized into a set of bins, with each bin assigned a Weight of Evidence (WoE), which enters the corresponding logistic scoring model.[14] The resulting contributions are subsequently rescaled so that the final PIT score ranges from 0 to 100. I observe the contribution of each financial characteristic to the final score,[15] with the individual contributions summing to the overall PIT score.

I first examine the persistence of these components. For each financial characteristic k and horizon h, I estimate

$$C_{i,t}^{k} = \alpha_{i}^{k,h} + \rho_{h}^{k} C_{i,t-h}^{k} + u_{i,t}^{k,h},$$

where $C_{i,t}^{k}$ denotes the contribution of characteristic k to firm i's PIT score. The regressions are estimated separately by firm-size category, and the fixed-effects estimates are corrected for short-panel bias using the half-panel jackknife of Dhaene and Jochmans (2015).[16] Figure 4 reports the estimated ρ_{h}^{k} for horizons from one to five years.

The figure reveals substantial differences in persistence across the financial characteristics entering the score. Financial Structure is consistently the most persistent component across firm sizes. Its one-year persistence is around 0.6–0.7 for large firms and remains above 0.5 for medium, small, and micro firms; moreover, its persistence declines only gradually as the horizon increases. Conversely, profitability and firm size display substantially less persistent within-firm variation for medium, small, and micro firms. For these groups, their one-year persistence is generally below 0.3 and falls rapidly toward zero as the horizon increases.

Large firms display a notably different pattern. Profitability is substantially more persistent and, more generally, the components entering their scores tend to evolve more persistently

[14] The Weight of Evidence measures how informative a given bin is about default risk. For a bin b, it is typically defined as

$$WoE_{b} = \log\left[\frac{\Pr(b \mid D=0)}{\Pr(b \mid D=1)}\right],$$

although the sign convention may be reversed.

[15] Some characteristics aggregate the contributions of several underlying financial ratios.

[16] The corrected estimate is $\tilde{\rho}_{h}^{k} = 2\hat{\rho}_{h}^{k} - \frac{1}{2}\left(\hat{\rho}_{h,1}^{k} + \hat{\rho}_{h,2}^{k}\right)$, where $\hat{\rho}_{h}^{k}$ is the fixed-effects estimate on the full sample and $\hat{\rho}_{h,1}^{k}$ and $\hat{\rho}_{h,2}^{k}$ are the estimates on its two halves. Since the panel is unbalanced, I follow Dhaene and Jochmans (2015) and treat it as a union of balanced panels: each firm's series is divided into spells of consecutive years, each with its own fixed effect, and every spell is split at its own midpoint. Spells with an odd number of years are split both ways and the resulting half-panel estimates averaged. Figure A.2 in the Appendix A reports the estimates obtained when the two halves are instead defined in calendar time. One-year persistence and the ranking of components are virtually unchanged, while persistence at longer horizons is somewhat higher.

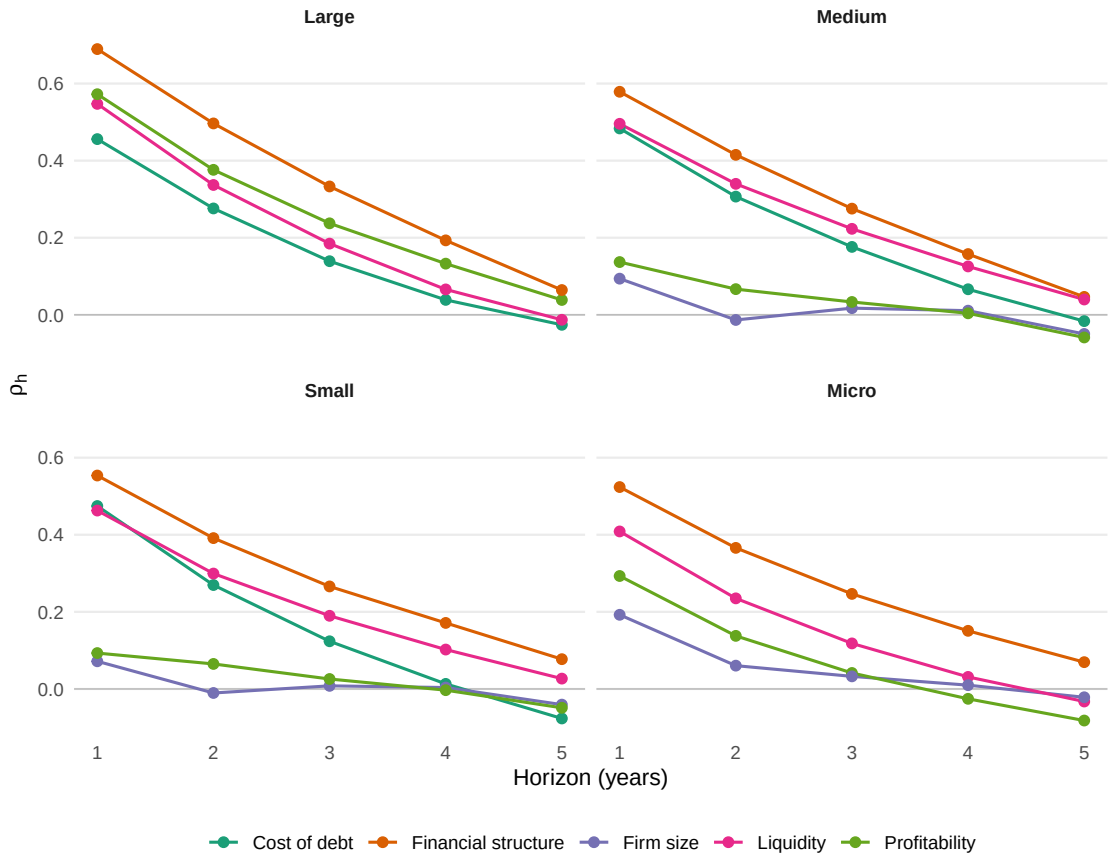


Figure 4: The figure reports estimated coefficients from regressions of each financial-score component on its lagged value at horizons of one to five years, controlling for firm fixed effects. Estimates are obtained separately by firm-size category and corrected for short-panel bias using the half-panel jackknife of Dhaene and Jochmans (2015).

than for smaller firms. This cross-size pattern is consistent with the results in Section 5: there is less short-lived firm-specific variation for filtering to remove among large firms, helping to explain both their smaller PIT–TTC reclassification rates and the limited gains from filtered TTC ratings.

As a complementary exercise, Appendix B examines the association between the individual financial components and the gap between PIT and TTC scores, controlling for firm and year fixed effects. Profitability and, where applicable, firm size generally exhibit stronger conditional associations with this gap than financial structure, consistent with their lower persistence. These regressions provide additional descriptive evidence, although their coefficients cannot be interpreted as contributions to transitory variation.

7 Heterogeneous cyclicality

The results in Section 6 suggest that firm-specific transitory variation plays an important role in explaining both the differences between PIT and TTC classifications and the gains in long-horizon risk differentiation from filtering. However, this interpretation relies on the baseline model's treatment of aggregate conditions. As illustrated in Section 2, firms may differ in their exposure to the business cycle, potentially affecting their relative rankings even without changes in persistent creditworthiness. If omitted, such heterogeneity could be attributed to persistent or firm-specific transitory variation, affecting both the smoothed TTC benchmark and the filtered ratings.

To examine this possibility, I extend the model to allow cyclical sensitivity to vary across industry–size groups and across firms within each group. I then assess whether this additional flexibility changes the smoothed TTC benchmark used to quantify PIT–TTC differences or improves the filtered TTC ratings constructed using current and past credit scores.

7.1 Modeling heterogeneous cyclicality

Using real GDP growth, X_t, to summarize aggregate conditions, I replace equation 5 with

$$y_{it} = \eta_{it} + \delta_{t,s_{it}} + \left(\overline{\lambda}_{s_{it},\iota_{it}} + \sigma_{s_{it},\iota_{it}}\lambda_i\right) X_t + \xi_{it}, \tag{11}$$

where ι_{it} denotes firm i's industry. The parameter $\overline{\lambda}_{s,\iota}$ captures average cyclical exposure in each industry–size group, while $\sigma_{s,\iota}\lambda_i$ captures firm-specific deviations. Because the year–size effects absorb cyclical movements common to firms of a given size, I normalize $\overline{\lambda}_{s,10} = 0$, with industry 10 corresponding to *Metallurgy and manufacture of metal products, except machinery and equipment* (henceforth, Metallurgy). Positive values of $\overline{\lambda}_{s,\iota}$ thus indicate greater average procyclicality than in Metallurgy firms of the same size.

Given the short histories of many firms, I follow an empirical Bayes approach[17] rather than estimating each loading separately. Specifically, I treat λ_i as a latent random effect with $\lambda_i \sim N(0,1)$, so that $\sigma_{s,\iota}$ measures the dispersion of cyclical exposure within each industry–size group. Although the loading is time-invariant, its contribution to the score varies with GDP growth.

[17] See Koenker and Gu (n.d.) and Koenker and Gu (2026) for an introduction to empirical Bayes methods.

The model can again be expressed in state-space form:

$$y_{it} = \begin{pmatrix} 1 & \sigma_{s_{it},\iota_{it}} X_t \end{pmatrix} \begin{pmatrix} \eta_{it} \\ \lambda_i \end{pmatrix} + \delta_{t,s_{it}} + \overline{\lambda}_{s_{it},\iota_{it}} X_t + \xi_{it}, \tag{12}$$

$$\begin{pmatrix} \eta_{it} \\ \lambda_i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \eta_{i,t-1} \\ \lambda_i \end{pmatrix} + \begin{pmatrix} \nu_{it} \\ 0 \end{pmatrix}. \tag{13}$$

Relative to the baseline, the latent state now includes the firm-specific cyclical loading. I estimate the model by maximum likelihood using the same EM algorithm and Kalman filtering and smoothing methods as in Section 4.

The resulting TTC log-odds measure evaluates aggregate conditions at their sample averages:

$$y_{it}^{HetTTC} = \mathbb{E}[\eta_{it} \mid \mathcal{I}_\tau] + \bar{\delta}_{s_{it}} + \left(\overline{\lambda}_{s_{it},\iota_{it}} + \sigma_{s_{it},\iota_{it}} \mathbb{E}[\lambda_i \mid \mathcal{I}_\tau]\right) \bar{X} + \beta_{it}, \tag{14}$$

where $\bar{X}$ is average GDP growth and $\tau \in \{t, T\}$. As before, $\tau = T$ gives the retrospective smoothed benchmark, whereas $\tau = t$ produces filtered ratings using information available up to the rating date, conditional on the estimated parameters. Both measures use the original rating thresholds.

7.2 Empirical implications

Table 9 shows that allowing for heterogeneous cyclicality changes only 2.5% of baseline filtered TTC ratings and 2.2% of smoothed TTC ratings. Differences are somewhat larger among large and medium firms but remain limited across sizes. Thus, neither the retrospective benchmark used to quantify PIT–TTC disagreement nor the feasible filtered classifications change materially.

TTC	All firms	Large	Medium	Small	Micro
Filtered	2.5%	5.3%	4.4%	3.8%	2.2%
Smoothed	2.2%	4.7%	4.0%	3.5%	2.0%

Table 9: Differences between baseline and heterogeneous-cyclicality TTC ratings. Entries report the share of observations assigned to different grades, separately for filtered and smoothed ratings.

Table 10 shows that risk differentiation is similarly unaffected. At horizons of one, three, six, and ten years, the full-sample AUROC differences relative to the baseline lie between -0.001 and $+0.001$ for both filtered and smoothed ratings. There is no systematic improvement across firm sizes: allowing for heterogeneous cyclicality neither increases the long-horizon gains from filtering for medium, small, and micro firms nor improves its performance for large

firms. The conclusion also holds among observations where PIT and smoothed TTC ratings disagree (Table A.3 in Appendix A).

Ratings	Within 1 year					Within 3 years				
	All firms	Large	Medium	Small	Micro	All firms	Large	Medium	Small	Micro
TTC f.	0.761	0.700	0.811	0.833	0.751	0.727	0.684	0.787	0.804	0.717
Het. TTC f.	-0.001	+0.001	-0.001	-0.001	0.000	-0.001	0.000	0.000	0.000	-0.001
TTC s.	0.776	0.710	0.824	0.846	0.766	0.760	0.704	0.821	0.834	0.749
Het. TTC s.	0.000	0.000	0.000	0.000	0.000	-0.001	-0.001	-0.001	0.000	0.000
Ratings	Within 6 years					Within 10 years				
	All firms	Large	Medium	Small	Micro	All firms	Large	Medium	Small	Micro
TTC f.	0.708	0.654	0.763	0.782	0.701	0.702	0.630	0.739	0.763	0.699
Het. TTC f.	0.000	0.000	0.000	0.000	0.000	-0.001	0.000	-0.001	0.000	-0.001
TTC s.	0.747	0.677	0.802	0.816	0.740	0.742	0.649	0.779	0.798	0.740
Het. TTC s.	-0.001	-0.001	-0.001	0.000	0.000	-0.001	-0.001	-0.001	0.000	-0.001

Table 10: Risk differentiation, measured by the AUROC, at horizons of 1, 3, 6, and 10 years. Baseline rows report AUROC levels; heterogeneous-cyclicality rows report differences relative to the corresponding baseline TTC ratings. Larger values indicate better risk differentiation.

The additional flexibility also has little effect on rating persistence: mobility measures and transition probabilities remain nearly unchanged (Tables A.4 and A.5 in Appendix A). Likewise, the estimated standard deviations of persistent innovations and transitory shocks remain close to the baseline estimates (Table A.6), preserving the greater relative importance of transitory variation among smaller firms.

Figure A.1 in Appendix A summarizes the estimated cyclical exposures. Most average industry loadings, measured relative to Metallurgy, are concentrated near zero. Within-industry dispersion is also relatively modest for most groups, although there are some substantial outliers. Thus, heterogeneous exposure is present in the estimates, but accounting for it has little effect on the resulting TTC classifications.

8 Conclusion

This paper has examined whether credit ratings designed to capture current default risk provide appropriate classifications of long-term creditworthiness. Using Bank of Spain credit scores for Spanish firms, I construct a retrospective benchmark based on smoothed estimates of persistent creditworthiness. PIT ratings differ from this benchmark in 51.2% of firm-year observations, with larger differences among smaller firms. Smoothed TTC ratings also outperform PIT ratings in differentiating future defaults, particularly at longer horizons, supporting the economic relevance of these discrepancies.

To mitigate these differences, I construct an alternative classification based on filtered estimates of persistent creditworthiness, using only current and past credit scores. Filtered TTC ratings reduce disagreement with the smoothed benchmark to 27.9%. More importantly, for medium, small, and micro firms, filtered TTC ratings increasingly outperform PIT ratings in differentiating future defaults as the horizon lengthens, while offering similar performance at shorter horizons. For large firms, by contrast, filtering does not improve risk differentiation at any horizon.

The results point to firm-specific transitory variation as an important source of PIT–TTC discrepancies and the gains from filtering, particularly among smaller firms. Such variation is relatively less important for large firms, whose credit scores and underlying financial components exhibit greater persistence. Neither removing aggregate fluctuations nor mechanically smoothing PIT scores reproduces the gains from filtering, while allowing for heterogeneous cyclical exposure to real GDP growth leaves the conclusions largely unchanged.

More broadly, the findings suggest that incorporating long-term creditworthiness exclusively through the calibration of default probabilities may be insufficient, as doing so does not address differences in borrower classifications. However, the benefits of adjusting borrower classifications depend on the importance of transitory variation in the underlying credit scores. Long-term rating design should therefore account not only for the intended horizon of the rating system, but also for the dynamics of the information used to differentiate borrowers.

Future research could extend the framework to distinguish structural changes in average creditworthiness from business-cycle fluctuations, particularly to investigate the sustained increase in the estimated year effects for micro firms. Another extension could compare filtered TTC ratings with ratings obtained from models estimated directly to predict defaults over longer horizons. Assessing their relative performance across horizons could help clarify the trade-offs associated with maintaining consistency between ratings used for risk management and regulatory capital purposes.

References

Aguais, S., Forest, L. R. J., King, M., Lennon, M. C., and Lordkipanidze, B. (2006). Designing and implementing a basel ii compliant pit–ttc ratings framework. In M. Ong (Ed.), *The basel handbook: A guide for financial practitioners* (2nd ed.). London, UK: Risk Books.

Aguais, S., Forest, L. R. J., Wong, E. Y. L., and Diaz-Ledezma, D. (2004). Point in time versus through the cycle ratings. In M. Ong (Ed.), *The basel handbook: A guide for financial practitioners.* London, UK: Risk Books.

Ahnert, T., Chapman, J., and Wilkins, C. (2021). Should bank capital regulation be risk sensitive? *Journal of financial intermediation*, *46*, 100870-.

Altman, E. I., and Rijken, H. A. (2004). How rating agencies achieve rating stability. *Journal of Banking & Finance*, *28*(11), 2679-2714.

Auria, L., Bingmer, M., Caicedo Graciano, C. M., Charavel, C., Gavilá, S., Iannamorelli, A., ... Sauer, S. (2021). *Overview of central banks' in-house credit assessment systems in the euro area* (Occasional Paper Series No. 284). European Central Bank. (Also Banco de España Occasional Paper 2131)

Basel Committee on Banking Supervision. (2006). *International convergence of capital measurement and capital standards: A revised framework.* Basel. (Basel II framework, see paragraph 444 (use of internal ratings))

Basel Committee on Banking Supervision. (2019). *Basel Framework: CRE36 – IRB Approach: Minimum Requirements to Use the IRB Approach.* Consolidated Basel Framework, Bank for International Settlements. Retrieved from https://www.bis.org/committees/bcbs/basel-framework/standard/cre/36/inforce/2019-12-15/published/2019-12-15 (See paragraphs CRE36.29–CRE36.30)

Behn, M., Haselmann, R., and Vig, V. (2022). The limits of model-based regulation. *The Journal of finance (New York)*, *77*(3), 1635-1684.

Behn, M., Haselmann, R., and Vig, V. (2025). Model-based capital regulation: Where do we stand and where should we go from here? [Journal Article]. *Annual Review of Financial Economics*, *17*(Volume 17, 2025), 93-111. Retrieved from https://www.annualreviews.org/content/journals/10.1146/annurev-financial-082123-110117 doi: https://doi.org/10.1146/annurev-financial-082123-110117

Berg, T., and Koziol, P. (2017). An analysis of the consistency of banks' internal ratings.

Journal of banking & finance, *78*, 27-41.

Carey, M., and Hrycay, M. (2001). Parameterizing credit risk models with rating data. *Journal of Banking & Finance*, *25*(1), 197-270. Retrieved from https://www.sciencedirect.com/science/article/pii/S0378426600001242 doi: https://doi.org/10.1016/S0378-4266(00)00124-2

Cesaroni, T. (2015). Procyclicality of credit rating systems: How to manage it. *Journal of Economics and Business*, *82*, 62-83. Retrieved from https://www.sciencedirect.com/science/article/pii/S0148619515000491 doi: https://doi.org/10.1016/j.jeconbus.2015.09.001

Dhaene, G., and Jochmans, K. (2015). Split-panel jackknife estimation of fixed-effect models. *The Review of economic studies*, *82*(3 (292)), 991-1030.

European Banking Authority. (2017a, November). *EBA Report on IRB Modelling Practices: Impact Assessment for the Guidelines on PD, LGD and the Treatment of Defaulted Exposures Based on the IRB Survey Results* (Tech. Rep.). European Banking Authority. Retrieved from https://www.eba.europa.eu/documents/10180/1720738/0212ecde-426d-4e18-84f8-04b036dcce00/EBA%20Report%20on%20IRB%20modelling%20practices.pdf

European Banking Authority. (2017b). *Guidelines on pd estimation, lgd estimation and the treatment of defaulted exposures.* (EBA/GL/2017/16)

European Banking Authority. (2023). *Supervisory Handbook on the Validation of Rating Systems under the Internal Ratings Based Approach* (Tech. Rep. No. EBA/REP/2023/29). European Banking Authority. Retrieved from https://www.eba.europa.eu/activities/single-rulebook/regulatory-activities/model-validation/supervisory-handbook-validation

European Parliament and Council of the European Union. (2013). *Regulation (EU) No 575/2013 of the European Parliament and of the Council of 26 June 2013 on prudential requirements for credit institutions and investment firms and amending Regulation (EU) No 648/2012.* Official Journal of the European Union, L 176, 27 June 2013, pp. 1–337. Retrieved from https://eur-lex.europa.eu/eli/reg/2013/575/oj

European Parliament and Council of the European Union. (2024). *Regulation (EU) 2024/1623 of the European Parliament and of the Council of 31 May 2024 amending Regulation (EU) No 575/2013 as regards requirements for credit risk, credit valuation adjustment risk, operational risk, market risk and the output floor.* Official Journal of the European

Union, L 2024/1623, 19 June 2024. Retrieved from https://eur-lex.europa.eu/eli/reg/2024/1623/oj

Fidalgo, O., Martínez, V., Moral Naranjo, E., and Oroz, M. (2024). A method for reducing credit scores' sensitivity to economic conditions. *Revista de Estabilidad Financiera - Banco de España*. Retrieved from https://repositorio.bde.es/handle/123456789/36636 doi: 10.53479/36636

Gavilá, S., Maldonado, A., and Marcelo, A. (2020). The Banco de España in-house credit assessment system. *Financial Stability Review, Banco de España*(38).

Gordy, M. B., and Howells, B. (2006). Procyclicality in basel ii: Can we treat the disease without killing the patient? *Journal of Financial Intermediation*, *15*(3), 395-417. Retrieved from https://www.sciencedirect.com/science/article/pii/S1042957306000088 (Basel II: Accounting, Transparency and Bank Stability) doi: https://doi.org/10.1016/j.jfi.2005.12.002

Gredil, O. R., Kapadia, N., and Lee, J. H. (2022, October). On the information content of credit ratings and market-based measures of default risk. *Journal of Financial Economics*, *146*(1), 172–204. Retrieved 2025-01-28, from https://linkinghub.elsevier.com/retrieve/pii/S0304405X22001507 doi: 10.1016/j.jfineco.2022.07.005

Jafry, Y., and Schuermann, T. (2004). Measurement, estimation and comparison of credit migration matrices. *Journal of banking & finance*, *28*(11), 2603-2639.

Kiff, J., Kisser, M., and Schumacher, L. (2013). *Rating through-the-cycle: What does the concept imply for rating stability and accuracy?* (IMF Working Paper No. 13/64). International Monetary Fund.

Koenker, R., and Gu, J. (n.d.). Empirical bayes for the reluctant frequentist. *Journal of Political Economy Microeconomics*, *0*(ja). Retrieved from https://www.journals.uchicago.edu/doi/abs/10.1086/743475 doi: 10.1086/743475

Koenker, R., and Gu, J. (2026). *Empirical bayes: Some tools, rules, and duals*. Cambridge University Press.

Löffler, G. (2004). An anatomy of rating through the cycle. *Journal of Banking & Finance*, *28*(3), 695-720. Retrieved from https://www.sciencedirect.com/science/article/pii/S0378426603000414 doi: https://doi.org/10.1016/S0378-4266(03)00041-4

Löffler, G. (2013). Can rating agencies look through the cycle? *Review of quantitative finance and accounting*, *40*(4), 623-646.

Mariathasan, M., and Merrouche, O. (2014). The manipulation of basel risk-weights. *Journal*

of financial intermediation, *23*(3), 300-321.

Peña, R. (2025). *Permanent and transitory unobserved heterogeneity in credit ratings* (Master's thesis, CEMFI). Retrieved from https://rodrigopenaill.github.io/files/master-thesis.pdf

Plosser, M. C., and Santos, J. A. C. (2018). Banks' incentives and inconsistent risk models. *The Review of financial studies*, *31*(6), 2080-2112.

Repullo, R., Saurina, J., and Trucharte, C. (2010, October). Mitigating the pro-cyclicality of Basel II: REFORMING BASEL II. *Economic Policy*, *25*(64), 659–702. Retrieved 2025-03-04, from https://academic.oup.com/economicpolicy/article-lookup/doi/10.1111/j.1468-0327.2010.00252.x doi: 10.1111/j.1468-0327.2010.00252.x

Rubtsov, M., and Petrov, A. (2016). A point-in-time–through-the-cycle approach to rating assignment and probability of default calibration. *Journal of Risk Model Validation*, *10*(2), 83–112.

Shumway, R. H., and Stoffer, D. S. (1982). An approach to time series smoothing and forecasting using the em algorithm. *Journal of Time Series Analysis*, *3*(4), 253-264. Retrieved from https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9892.1982.tb00349.x doi: https://doi.org/10.1111/j.1467-9892.1982.tb00349.x

Treacy, W. F., and Carey, M. (2000). Credit risk rating systems at large us banks. *Journal of Banking & Finance*, *24*(1), 167-201. Retrieved from https://www.sciencedirect.com/science/article/pii/S0378426699000564 doi: https://doi.org/10.1016/S0378-4266(99)00056-4

Appendix A

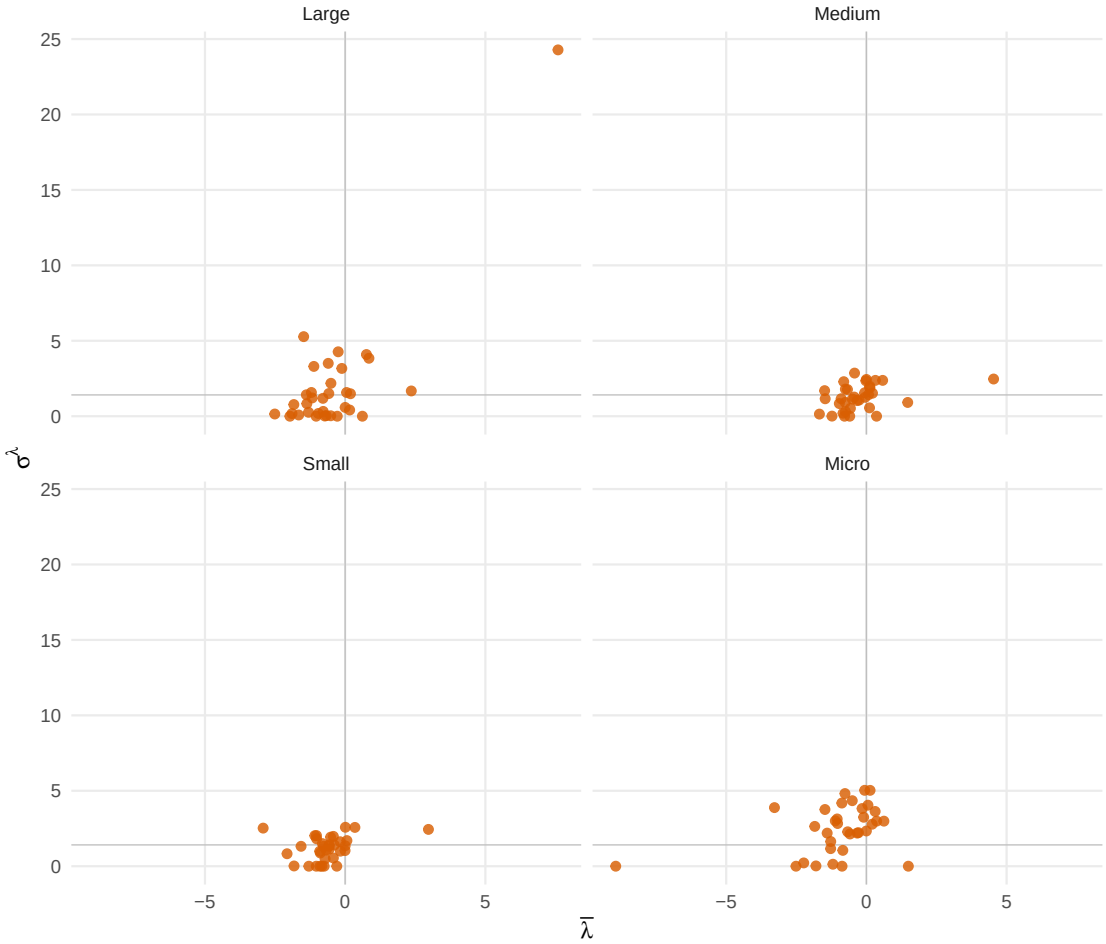


Figure A.1: Estimated mean and standard deviation of cyclical exposure by firm size and industry. Each point corresponds to an industry within a firm-size group. The horizontal axis reports the estimated average cyclical exposure relative to the Metallurgy sector, while the vertical axis reports the estimated standard deviation of firm-specific cyclical exposure within the industry–size group. Industries correspond to the 34 CNAE sector groups listed in table A.7 in the Appendix A.

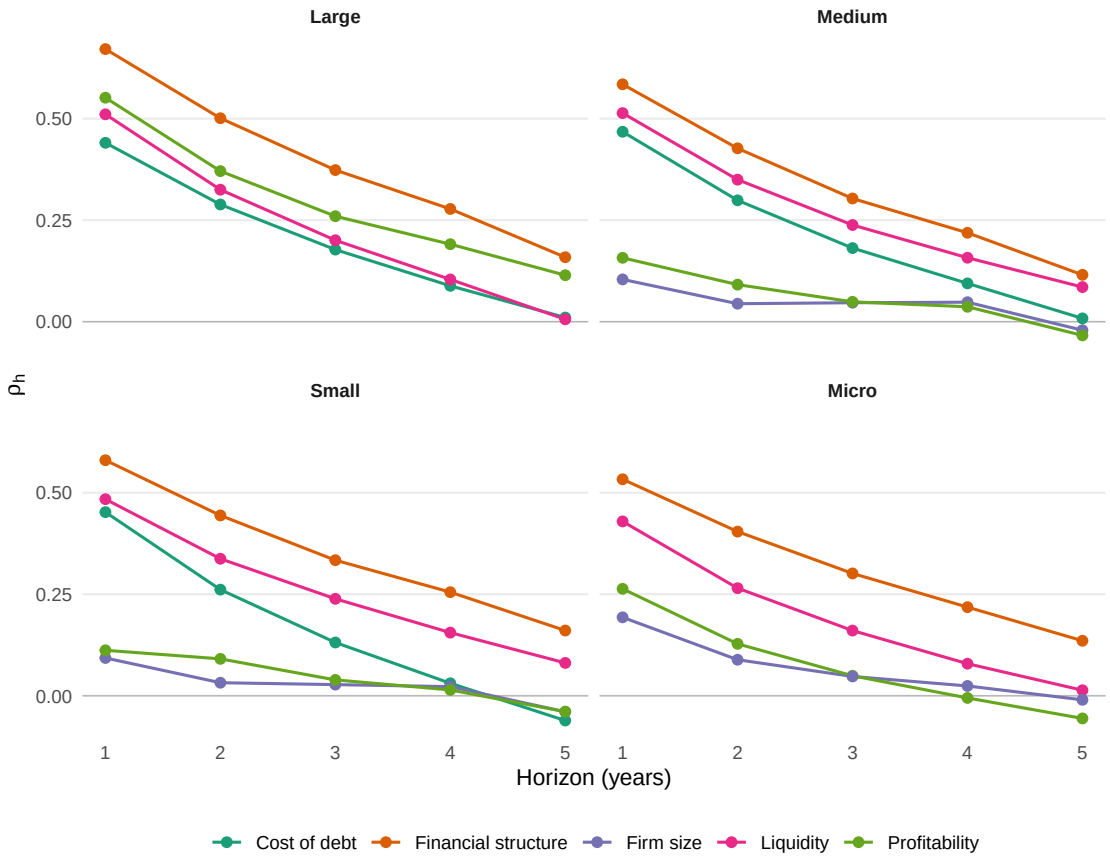


Figure A.2: Persistence of within-firm movements in the scoring-model components, with the half-panel jackknife halves defined in calendar time. The sample is split at the middle year of the sample period rather than at the midpoint of each firm's spell.

Sample	Subset	Within 1 year				Within 3 years			
		PIT	MA 3	TTC f.	TTC s.	PIT	MA 3	TTC f.	TTC s.
All firms	All	0.764	-0.019	-0.003	+0.012	0.727	-0.009	0.000	+0.033
	PIT ≠ TTC s.	0.716	-0.025	-0.002	+0.021	0.665	-0.006	+0.008	+0.059
	PIT ≠ TTC f.	0.727	-0.033	-0.009	+0.012	0.691	-0.018	-0.004	+0.040
Large	All	0.706	-0.017	-0.006	+0.005	0.690	-0.016	-0.007	+0.014
	PIT ≠ TTC s.	0.709	-0.028	-0.007	+0.009	0.667	-0.015	-0.004	+0.032
	PIT ≠ TTC f.	0.719	-0.038	-0.014	+0.006	0.695	-0.032	-0.016	+0.015
Medium	All	0.809	-0.011	+0.002	+0.015	0.787	-0.007	0.000	+0.034
	PIT ≠ TTC s.	0.744	-0.016	+0.006	+0.035	0.695	+0.003	+0.012	+0.076
	PIT ≠ TTC f.	0.757	-0.024	+0.002	+0.025	0.731	-0.015	-0.004	+0.049
Small	All	0.831	-0.013	+0.003	+0.015	0.800	-0.008	+0.003	+0.034
	PIT ≠ TTC s.	0.750	-0.014	+0.009	+0.034	0.702	+0.004	+0.019	+0.076
	PIT ≠ TTC f.	0.766	-0.020	+0.004	+0.024	0.735	-0.010	+0.004	+0.052
Micro	All	0.753	-0.013	-0.002	+0.013	0.716	-0.003	+0.001	+0.033
	PIT ≠ TTC s.	0.714	-0.020	-0.001	+0.021	0.664	-0.001	+0.009	+0.057
	PIT ≠ TTC f.	0.724	-0.027	-0.006	+0.014	0.687	-0.011	0.000	+0.040
Sample	Subset	Within 6 years				Within 10 years			
		PIT	MA 3	TTC f.	TTC s.	PIT	MA 3	TTC f.	TTC s.
All firms	All	0.705	-0.003	+0.004	+0.042	0.693	0.000	+0.009	+0.048
	PIT ≠ TTC s.	0.638	+0.007	+0.017	+0.076	0.629	+0.013	+0.024	+0.085
	PIT ≠ TTC f.	0.667	-0.006	+0.004	+0.053	0.655	+0.001	+0.013	+0.062
Large	All	0.660	-0.016	-0.005	+0.017	0.630	-0.012	-0.001	+0.019
	PIT ≠ TTC s.	0.626	-0.009	+0.001	+0.041	0.596	+0.001	+0.011	+0.044
	PIT ≠ TTC f.	0.657	-0.026	-0.014	+0.020	0.618	-0.013	-0.001	+0.030
Medium	All	0.761	-0.006	+0.001	+0.041	0.730	-0.004	+0.008	+0.049
	PIT ≠ TTC s.	0.659	+0.011	+0.019	+0.090	0.627	+0.018	+0.032	+0.104
	PIT ≠ TTC f.	0.702	-0.009	0.000	+0.060	0.667	-0.001	+0.017	+0.076
Small	All	0.773	-0.005	+0.008	+0.043	0.747	-0.001	+0.016	+0.051
	PIT ≠ TTC s.	0.665	+0.017	+0.032	+0.096	0.642	+0.026	+0.045	+0.107
	PIT ≠ TTC f.	0.700	+0.001	+0.017	+0.069	0.671	+0.014	+0.034	+0.083
Micro	All	0.698	+0.003	+0.003	+0.042	0.692	+0.006	+0.007	+0.048
	PIT ≠ TTC s.	0.643	+0.011	+0.015	+0.072	0.639	+0.016	+0.020	+0.080
	PIT ≠ TTC f.	0.670	0.000	+0.005	+0.051	0.663	+0.005	+0.011	+0.059

Table A.1: The table reports PIT AUROC values and differences relative to PIT for MA 3, filtered TTC, and smoothed TTC ratings. Results are presented for the full sample and for observations in which PIT ratings differ from smoothed or filtered TTC ratings.

Sample	Subset	Within 1 year				Within 3 years			
		PIT	PIT adj.	TTC f.	TTC s.	PIT	PIT adj.	TTC f.	TTC s.
All firms	All	0.764	-0.003	-0.003	+0.012	0.727	-0.005	0.000	+0.033
	PIT ≠ TTC s.	0.716	-0.001	-0.002	+0.021	0.665	-0.001	+0.008	+0.059
	PIT ≠ TTC f.	0.727	-0.003	-0.009	+0.012	0.691	-0.004	-0.004	+0.040
Large	All	0.706	-0.002	-0.006	+0.005	0.690	-0.002	-0.007	+0.014
	PIT ≠ TTC s.	0.709	-0.002	-0.007	+0.009	0.667	0.000	-0.004	+0.032
	PIT ≠ TTC f.	0.719	-0.005	-0.014	+0.006	0.695	-0.005	-0.016	+0.015
Medium	All	0.809	+0.001	+0.002	+0.015	0.787	-0.003	0.000	+0.034
	PIT ≠ TTC s.	0.744	+0.006	+0.006	+0.035	0.695	+0.005	+0.012	+0.076
	PIT ≠ TTC f.	0.757	+0.005	+0.002	+0.025	0.731	-0.001	-0.004	+0.049
Small	All	0.831	0.000	+0.003	+0.015	0.800	-0.003	+0.003	+0.034
	PIT ≠ TTC s.	0.750	+0.005	+0.009	+0.034	0.702	+0.005	+0.019	+0.076
	PIT ≠ TTC f.	0.766	+0.003	+0.004	+0.024	0.735	0.000	+0.004	+0.052
Micro	All	0.753	-0.002	-0.002	+0.013	0.716	-0.003	+0.001	+0.033
	PIT ≠ TTC s.	0.714	+0.001	-0.001	+0.021	0.664	0.000	+0.009	+0.057
	PIT ≠ TTC f.	0.724	0.000	-0.006	+0.014	0.687	-0.001	0.000	+0.040

Sample	Subset	Within 6 years				Within 10 years			
		PIT	PIT adj.	TTC f.	TTC s.	PIT	PIT adj.	TTC f.	TTC s.
All firms	All	0.705	-0.003	+0.004	+0.042	0.693	+0.001	+0.009	+0.048
	PIT ≠ TTC s.	0.638	+0.002	+0.017	+0.076	0.629	+0.007	+0.024	+0.085
	PIT ≠ TTC f.	0.667	-0.002	+0.004	+0.053	0.655	+0.003	+0.013	+0.062
Large	All	0.660	-0.001	-0.005	+0.017	0.630	+0.002	-0.001	+0.019
	PIT ≠ TTC s.	0.626	+0.003	+0.001	+0.041	0.596	+0.008	+0.011	+0.044
	PIT ≠ TTC f.	0.657	-0.002	-0.014	+0.020	0.618	+0.004	-0.001	+0.030
Medium	All	0.761	-0.002	+0.001	+0.041	0.730	+0.004	+0.008	+0.049
	PIT ≠ TTC s.	0.659	+0.007	+0.019	+0.090	0.627	+0.017	+0.032	+0.104
	PIT ≠ TTC f.	0.702	0.000	0.000	+0.060	0.667	+0.011	+0.017	+0.076
Small	All	0.773	+0.001	+0.008	+0.043	0.747	+0.008	+0.016	+0.051
	PIT ≠ TTC s.	0.665	+0.013	+0.032	+0.096	0.642	+0.023	+0.045	+0.107
	PIT ≠ TTC f.	0.700	+0.006	+0.017	+0.069	0.671	+0.017	+0.034	+0.083
Micro	All	0.698	-0.002	+0.003	+0.042	0.692	0.000	+0.007	+0.048
	PIT ≠ TTC s.	0.643	+0.001	+0.015	+0.072	0.639	+0.004	+0.020	+0.080
	PIT ≠ TTC f.	0.670	-0.001	+0.005	+0.051	0.663	+0.003	+0.011	+0.059

Table A.2: The table reports PIT AUROC values and differences relative to PIT for aggregate-adjusted PIT, filtered TTC, and smoothed TTC ratings. Results are presented for the full sample and for observations in which PIT ratings differ from smoothed or filtered TTC ratings.

Ratings	Within 1 year					Within 3 years				
	All firms	Large	Medium	Small	Micro	All firms	Large	Medium	Small	Micro
TTC f.	0.714	0.702	0.750	0.759	0.713	0.673	0.662	0.708	0.721	0.673
Het. TTC f.	-0.001	+0.002	-0.001	-0.001	0.000	-0.001	+0.001	-0.001	0.000	-0.001
TTC s.	0.737	0.718	0.778	0.785	0.734	0.724	0.699	0.771	0.778	0.721
Het. TTC s.	0.000	+0.002	-0.001	-0.001	0.000	-0.001	0.000	-0.002	-0.001	-0.001
Ratings	Within 6 years					Within 10 years				
	All firms	Large	Medium	Small	Micro	All firms	Large	Medium	Small	Micro
TTC f.	0.655	0.626	0.678	0.698	0.658	0.653	0.606	0.659	0.687	0.659
Het. TTC f.	0.000	+0.001	0.000	0.000	0.000	-0.001	-0.001	-0.001	-0.001	-0.001
TTC s.	0.714	0.667	0.748	0.761	0.715	0.713	0.640	0.731	0.749	0.719
Het. TTC s.	-0.001	-0.001	-0.002	-0.001	-0.001	-0.002	-0.001	-0.003	-0.001	-0.001

Table A.3: Risk differentiation, measured using the AUROC, across models and firm sizes, for 1,3,6, and 10 periods ahead. This is calculated in the subsample of observations where PIT and TTC smoothed ratings differ. Results for the models allowing for heterogeneous sensitivity to the business cycle are presented as the difference with respect to the baseline model. Larger values are associated with greater risk differentiation.

Order	Ratings	All firms	Large	Medium	Small	Micro
1st order	TTC f.	0.418	0.477	0.350	0.368	0.279
	Het. TTC f.	0.418	0.480	0.351	0.368	0.279
	TTC s.	0.296	0.415	0.266	0.289	0.185
	Het. TTC s.	0.297	0.418	0.268	0.290	0.185
2nd order	TTC f.	0.537	0.596	0.422	0.440	0.369
	Het. TTC f.	0.536	0.596	0.421	0.439	0.369
	TTC s.	0.442	0.563	0.372	0.394	0.290
	Het. TTC s.	0.443	0.567	0.372	0.395	0.290

Table A.4: Persistence measure proposed by Jafry and Schuermann (2004) for 1st and 2nd order transition matrices, for all firms and for each size category. Lower values are associated with greater persistence in the transition matrix.

Ratings	4+	4	4-	5+	5	5-	6+	6	6-	7+	7	7-	D
All firms													
TTC f.	0.60	0.84	0.66	0.78	0.58	0.33	0.69	0.37	0.50	0.50	0.71	0.33	0.67
Het. TTC f.	0.60	0.84	0.66	0.78	0.58	0.33	0.68	0.37	0.50	0.50	0.71	0.32	0.67
TTC s.	0.71	0.89	0.74	0.87	0.74	0.46	0.81	0.60	0.71	0.69	0.83	0.44	0.67
Het. TTC s.	0.71	0.89	0.74	0.87	0.74	0.46	0.81	0.60	0.71	0.69	0.83	0.44	0.67
Large													
TTC f.	0.60	0.73	0.66	0.58	0.49	0.61	0.37	0.24	0.31	0.49	0.39	0.40	0.71
Het. TTC f.	0.60	0.73	0.66	0.58	0.48	0.60	0.37	0.24	0.31	0.49	0.37	0.40	0.71
TTC s.	0.71	0.81	0.74	0.67	0.58	0.70	0.47	0.32	0.41	0.57	0.43	0.33	0.71
Het. TTC s.	0.71	0.80	0.74	0.67	0.57	0.69	0.46	0.32	0.40	0.57	0.42	0.33	0.71
Medium													
TTC f.	—	0.86	—	0.59	—	0.35	0.42	0.37	—	0.44	0.18	0.39	0.73
Het. TTC f.	—	0.86	—	0.59	—	0.35	0.42	0.37	—	0.44	0.18	0.39	0.73
TTC s.	—	0.92	—	0.73	—	0.51	0.59	0.54	—	0.62	0.31	0.50	0.73
Het. TTC s.	—	0.92	—	0.73	—	0.50	0.58	0.54	—	0.61	0.31	0.50	0.73
Small													
TTC f.	—	0.83	—	0.54	—	0.30	0.41	0.36	—	0.44	0.18	0.31	0.70
Het. TTC f.	—	0.83	—	0.54	—	0.30	0.41	0.36	—	0.44	0.18	0.31	0.70
TTC s.	—	0.89	—	0.68	—	0.44	0.56	0.51	—	0.59	0.31	0.43	0.70
Het. TTC s.	—	0.89	—	0.68	—	0.44	0.56	0.51	—	0.59	0.30	0.43	0.70
Micro													
TTC f.	—	—	—	0.84	0.58	—	0.70	0.37	0.50	0.52	0.75	0.34	0.69
Het. TTC f.	—	—	—	0.84	0.58	—	0.70	0.37	0.50	0.52	0.75	0.34	0.69
TTC s.	—	—	—	0.92	0.74	—	0.83	0.61	0.71	0.72	0.87	0.47	0.69
Het. TTC s.	—	—	—	0.92	0.74	—	0.83	0.61	0.71	0.72	0.87	0.47	0.69

Table A.5: The table reports diagonal elements of the one-year transition matrices for filtered and smoothed TTC ratings under the baseline and heterogeneous-cyclicality models. Entries represent the probability of remaining in the same rating category.

Parameter	Model	Large	Medium	Small	Micro
$\sigma_{0,s}$	Baseline	0.74	0.56	0.57	0.51
	Heterogeneous	0.74	0.56	0.57	0.50
$\sigma_{\eta,s}$	Baseline	0.35	0.29	0.32	0.42
	Heterogeneous	0.36	0.29	0.32	0.43
$\sigma_{\varepsilon,s}$	Baseline	0.39	0.41	0.43	0.86
	Heterogeneous	0.37	0.40	0.42	0.85
$\sigma_{\eta,s}/\sigma_{\varepsilon,s}$	Baseline	0.91	0.72	0.75	0.49
	Heterogeneous	0.95	0.74	0.76	0.50

Table A.6: Estimated standard deviation parameters by firm size.

Table A.7: Sectoral aggregation: Spanish original and English translation

#	**Agrupación sectorial**	**Sectoral aggregation**
1	Agricultura, ganadería, silvicultura y pesca	Agriculture, livestock, forestry and fishing
2	Petróleo (extracción y refino)	Petroleum: extraction and refining
3	Otras industrias extractivas	Other extractive industries
4	Alimentación, bebidas y tabaco	Food, beverages and tobacco
5	Textil, confección, cuero y calzado	Textiles, clothing, leather and footwear
6	Madera y corcho, papel y artes gráficas	Wood and cork, paper and graphic arts
7	Industria química y farmacéutica	Chemical and pharmaceutical industry
8	Caucho y materias plásticas	Rubber and plastic products
9	Productos minerales no metálicos diversos	Other non-metallic mineral products
10	Metalurgia y fabricación de productos metálicos, excepto maquinaria y equipo	Metallurgy and manufacture of metal products, except machinery and equipment
11	Electrónica	Electronics
12	Material y equipo eléctrico, electrónico y óptico	Electrical, electronic and optical equipment
13	Maquinaria y equipo mecánico	Mechanical machinery and equipment
14	Automoción	Automotive industry
15	Construcción aeronáutica	Aeronautical construction
16	Otro material de transporte	Other transport equipment
17	Industrias manufactureras diversas, reparación e instalación de maquinaria de equipo	Other manufacturing industries; repair and installation of machinery and equipment
18	Suministro de energía eléctrica, gas, vapor y aire acondicionado	Electricity, gas, steam and air conditioning supply
19	Suministro de agua, actividades de saneamiento, gestión de residuos y descontaminación	Water supply, sewerage, waste management and remediation activities

Continued on next page

#	**Agrupación sectorial**	**Sectoral aggregation**
20	Construcción	Construction
21	Otras actividades relacionadas con la construcción	Other construction-related activities
22	Venta y reparación de vehículos de motor y motocicletas	Sale and repair of motor vehicles and motorcycles
23	Comercio al por mayor e intermediarios del comercio, excepto de vehículos de motor y motocicletas	Wholesale trade and trade intermediaries, except motor vehicles and motorcycles
24	Comercio al por menor, excepto de vehículos de motor y motocicletas	Retail trade, except motor vehicles and motorcycles
25	Transporte y almacenamiento	Transportation and storage
26	Hostelería	Accommodation and food service activities
27	Información y comunicaciones	Information and communication
28	Actividades inmobiliarias	Real estate activities
29	Actividades financieras y de seguros	Financial and insurance activities
30	Actividades profesionales, científicas y técnicas	Professional, scientific and technical activities
31	Actividades administrativas y servicios auxiliares	Administrative and support service activities
32	Sanidad	Human health activities
33	Otros	Other activities
34	Otras actividades de construcción especializada	Other specialized construction activities

Appendix B

Section 6.2 documents substantial differences in the persistence of the financial components entering PIT scores. As a complementary exercise, I examine whether movements in these components are systematically associated with the gap between current and long-term creditworthiness. Specifically, for each firm-size category I estimate

$$PIT_{it} - TTC_{it} = \alpha_i + \gamma_t + \sum_k \beta_k C_{it}^k + \varepsilon_{it},$$

where C_{it}^k denotes the contribution of characteristic k to the PIT score. I estimate the specification separately using the filtered and smoothed TTC scores.

These regressions should be interpreted as descriptive rather than as a decomposition of the PIT–TTC gap. In particular, the score components jointly determine the PIT score and can move together within firms. Each coefficient measures how the PIT–TTC gap changes with a one-point movement in that component, conditional on the remaining components and firm and year fixed effects[18]. The coefficients are therefore neither causal effects nor shares of transitory variation.

Table B.1 reveals a similar pattern for both filtered and smoothed TTC scores. Profitability consistently exhibits the largest coefficient across firm sizes, while financial structure has the smallest. Firm size, where it enters the scoring model, is also strongly associated with the PIT–TTC gap. These findings complement the persistence evidence in Figure 4: components exhibiting less persistent within-firm movements tend to be more strongly associated with deviations between current and long-term creditworthiness.

[18] Indeed, since the contributions add up to the PIT score, these coefficients are equal to one minus the coefficient from projecting the TTC score on the same regressors.

	Gap to the filtered TTC score				Gap to the smoothed TTC score			
	Large (1)	Medium (2)	Small (3)	Micro (4)	Large (5)	Medium (6)	Small (7)	Micro (8)
Firm size	—	0.349***	0.310***	0.554***	—	0.598***	0.558***	0.711***
		(0.003)	(0.001)	(0.000)		(0.003)	(0.001)	(0.000)
Financial structure	0.182***	0.172***	0.156***	0.284***	0.260***	0.305***	0.297***	0.461***
	(0.003)	(0.002)	(0.001)	(0.000)	(0.004)	(0.002)	(0.001)	(0.000)
Cost of debt	0.216***	0.229***	0.277***	—	0.359***	0.395***	0.405***	—
	(0.005)	(0.004)	(0.002)		(0.006)	(0.004)	(0.002)	
Liquidity	0.214***	0.286***	0.256***	0.467***	0.308***	0.396***	0.357***	0.580***
	(0.008)	(0.003)	(0.001)	(0.000)	(0.009)	(0.003)	(0.001)	(0.000)
Profitability	0.349***	0.742***	0.776***	1.333***	0.412***	0.773***	0.788***	0.848***
	(0.010)	(0.004)	(0.002)	(0.002)	(0.012)	(0.004)	(0.002)	(0.002)
Observations	78,669	362,652	2,340,662	16,285,073	78,669	362,652	2,340,662	16,285,073
Firms	9,948	59,029	352,785	1,919,404	9,948	59,029	352,785	1,919,404
Within R^2	0.358	0.556	0.551	0.574	0.498	0.702	0.678	0.742

Table B.1: Association between financial-score components and the PIT–TTC gap. The dependent variable is the gap between the original PIT model score and the TTC score, so a positive coefficient is associated with a larger PIT–TTC score gap. All columns absorb firm and year fixed effects; standard errors, in parentheses, are clustered by firm. The underlying scoring model differs across large firms, medium/small firms, and micro firms. Hence, the set of components differs across columns: — marks a component that does not enter that model. ***, ** and * denote significance at the 1%, 5% and 10% level.